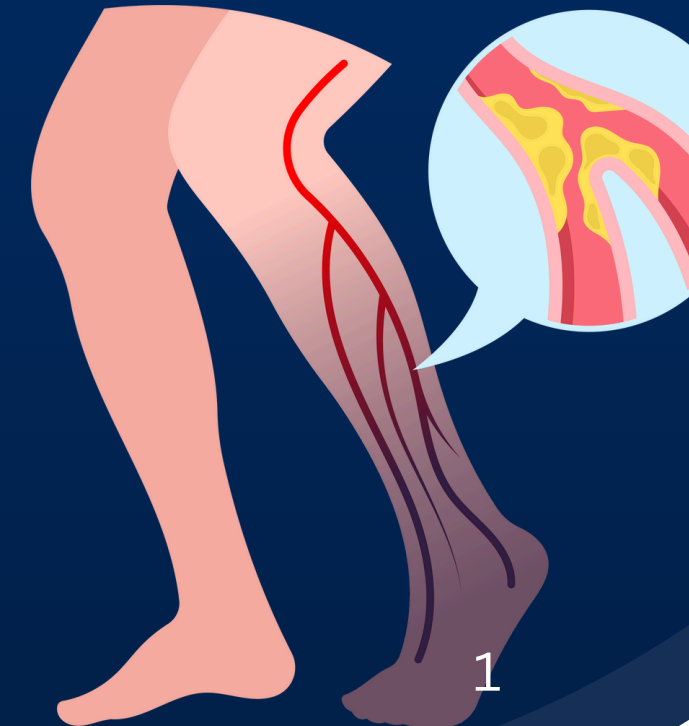
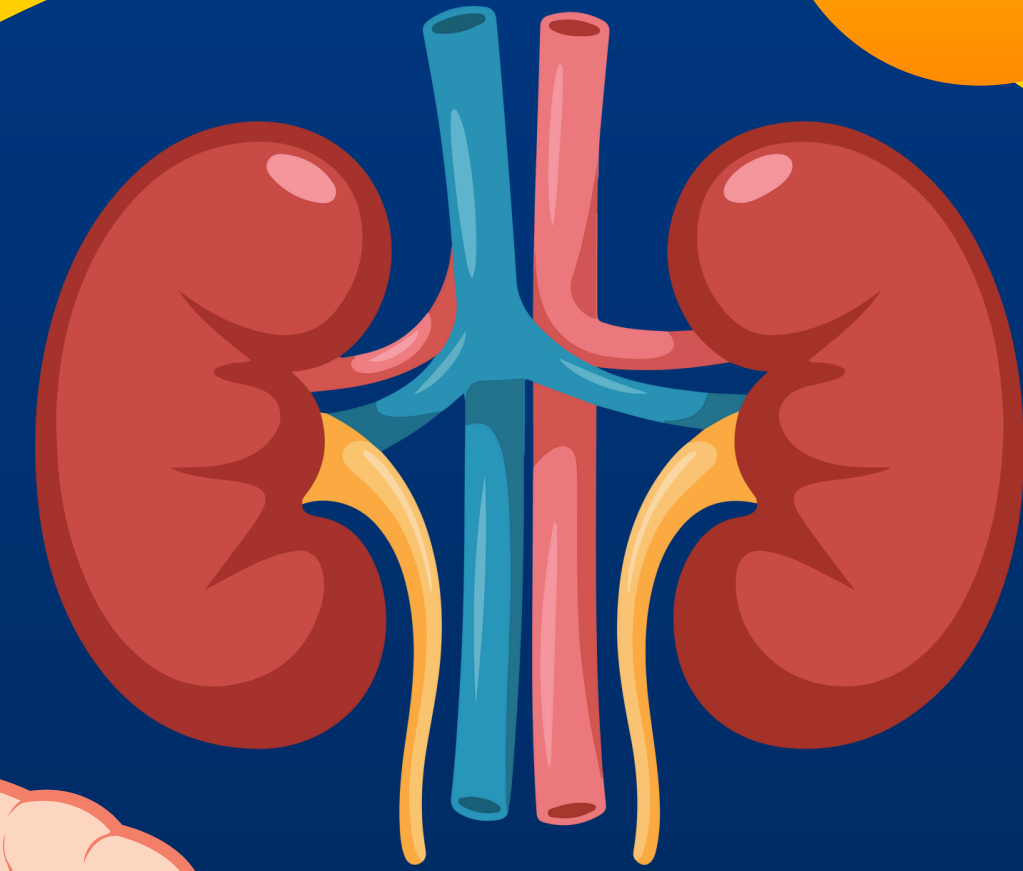
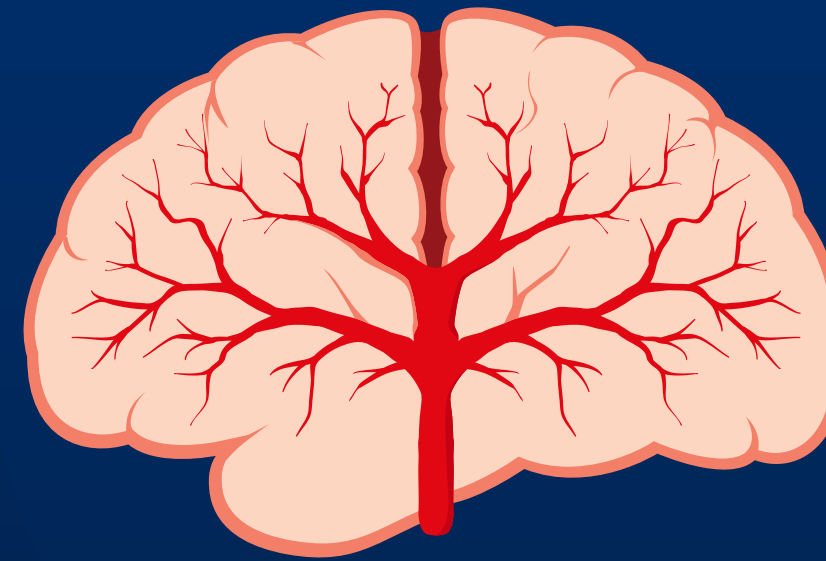
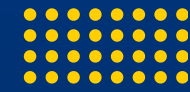


Circulation in specific vascular beds

Assist.Prof.Dr.Nifareeda Samerphob



Division of health and applied sciences,
Faculty of science, Prince of Songkla University



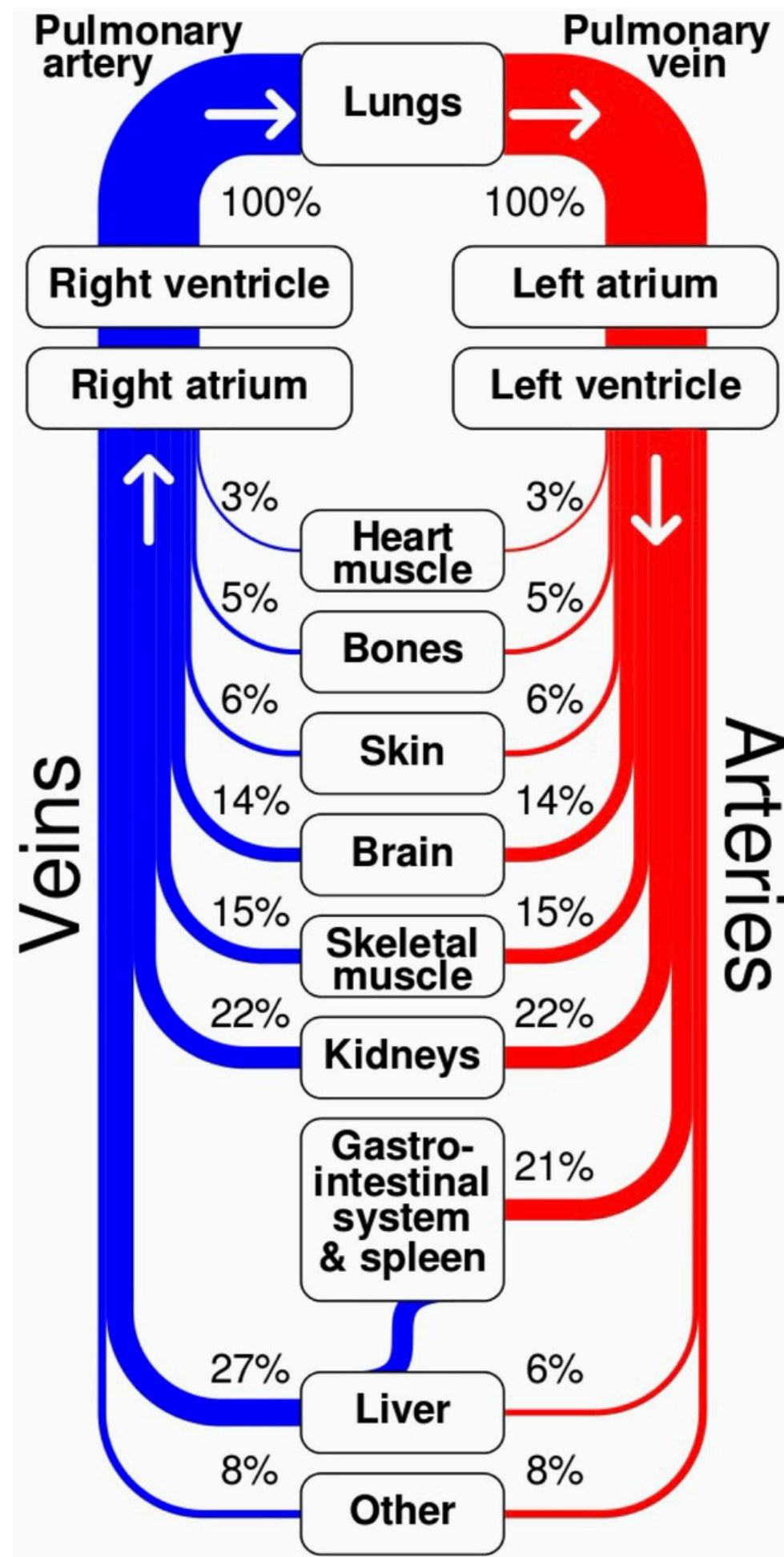
LEARNING OBJECTIVES

1.Principles of organ blood flow regulation

- 1.1 extrinsic control (nervous and chemical control)
- 1.2 intrinsic control (local regulation/autoregulation)

2. Regulation of circulation in specific vascular bed

- 2.1 Coronary circulation
- 2.2 Cerebral circulation
- 2.3 Skeletal muscle circulation
- 2.4 Cutaneous circulation
- 2.5 Splanchnic circulation
- 2.6 Renal circulation



- The heart functions in three aspects: as the right pump, the left pump, and as a muscle with **its own metabolic and flow demands**.
 - **Cardiac output rises** equally from both sides, so increased systemic demand automatically boosts pulmonary flow and oxygen delivery.
 - **Systemic organs are arranged in parallel**, ensuring identical arterial blood supply and **allowing independent flow regulation**.
- Systemic organs receive the same arterial blood while **each organ regulates its own blood flow independently**.

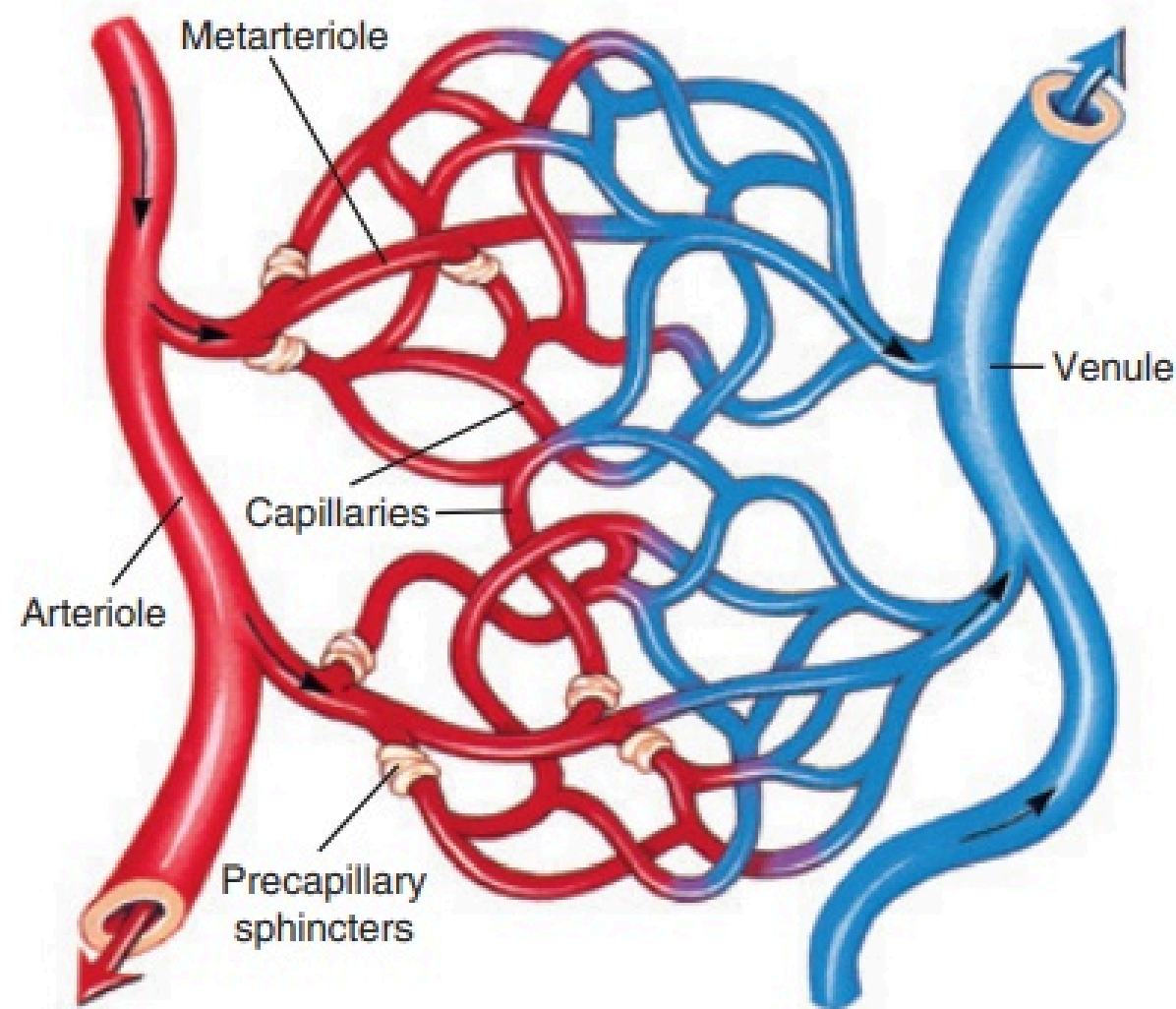


FIGURE 19-4 Just as we travel on different types of roads, our vasculature consists of several different types of vessels.

- Special circulations (coronary, cerebral, renal, hepatic, pulmonary) have unique mechanisms regulating blood flow beyond standard control systems.
- Blood flow depends on:
 - The **pressure gradient (driving force)**
 - **Vascular resistance**, determined by vessel radius, length, and blood viscosity
- Arteriolar radius has the greatest influence on flow.
 - Poiseuille's law (approximation):

$$Q = \frac{\pi(P_1 - P_2)r^4}{8\eta l}$$

Q : flow rate

$P_1 - P_2$: pressure gradient

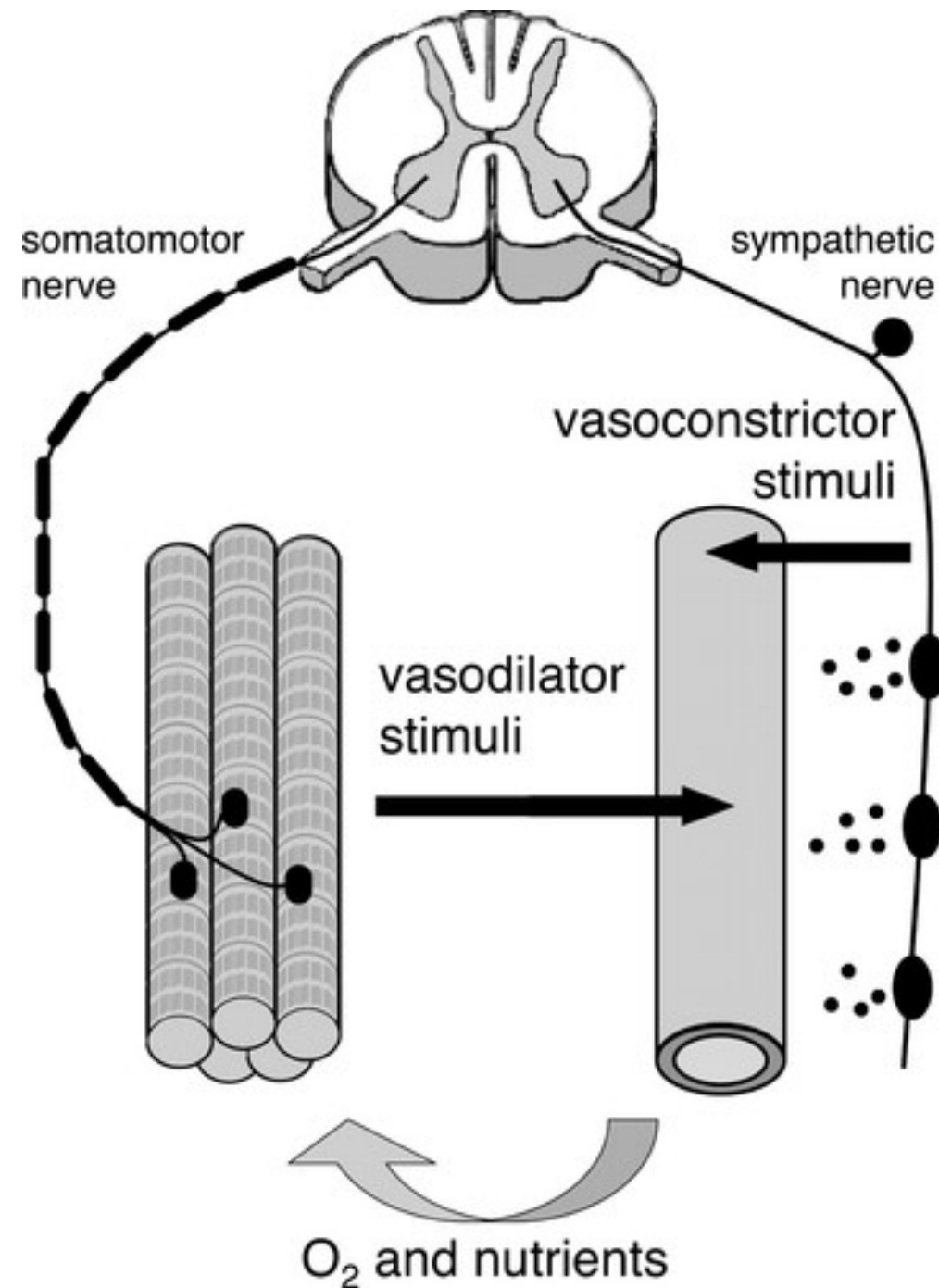
r : vessel radius

η : blood viscosity

l : vessel length

- Because flow doubling vessel radius increases flow 16-fold.
- Vascular tone is controlled by:
 - **Extrinsic factors: neural and humoral influences**
 - **Intrinsic factors: myogenic and metabolic mechanisms.**

Extrinsic control



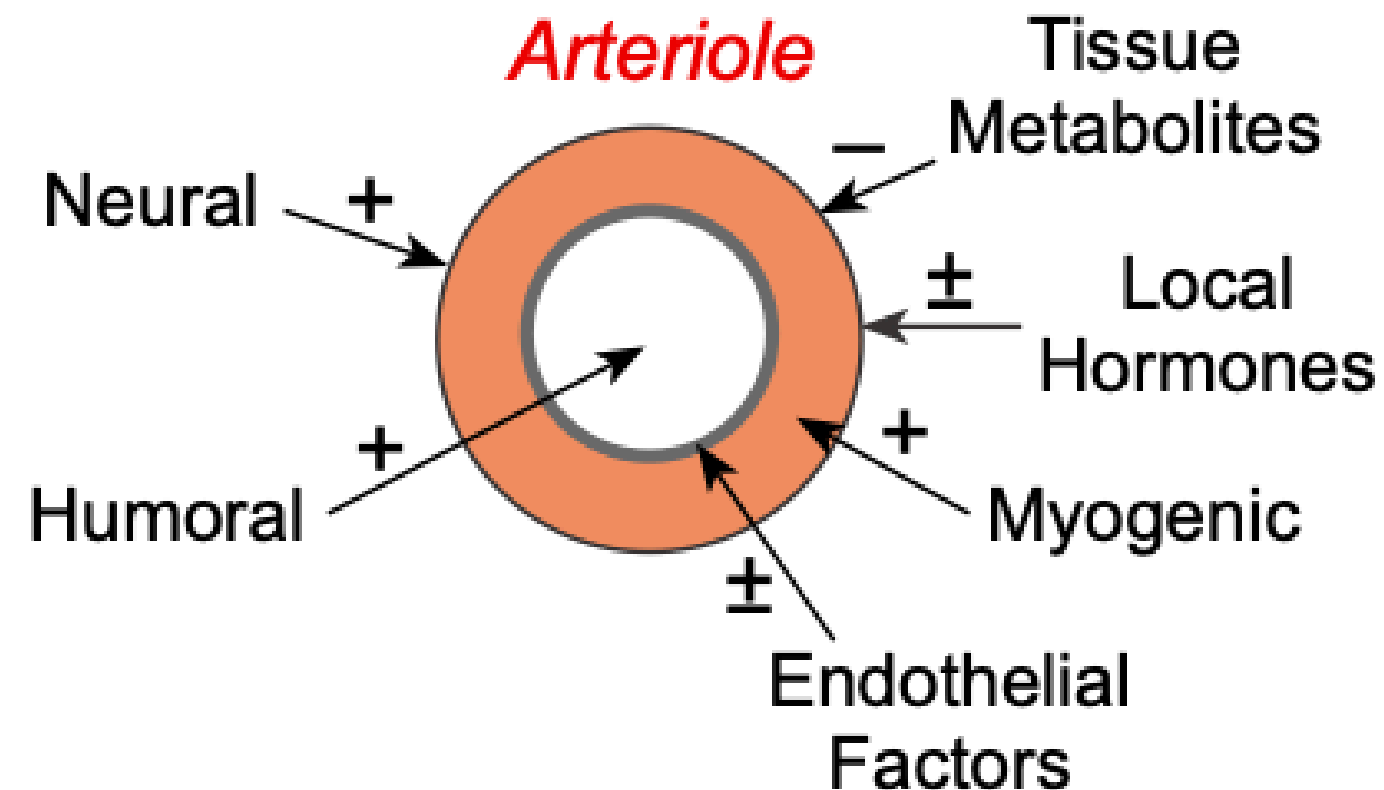
Neural control of muscle blood flow during exercise | Journal of Applied Physiology | American Physiological Society

- **Sympathetic innervation:** vasomotor control is mainly sympathetic; vasoconstrictor sympathetic fibers maintain basal vascular tone through tonic activity.
 - Neurotransmitter: Noradrenaline (norepinephrine) acts on α - and β -adrenoreceptors.
 - α -adrenoreceptor activation $\rightarrow$ vasoconstriction (dominant effect in most vascular beds).
- **Parasympathetic vasodilation occurs in limited regions** – cerebral and meningeal vessels, viscera, genitalia, large bowel, and bladder (not in skeletal muscle or skin).
 - Cutaneous vasodilation during sweating partly results from bradykinin released by exocrine glands.
- **Circulatory vasoactive substance:**
 - adrenaline effects depend on concentration:
 - Low levels: β -receptor activation $\rightarrow$ vasodilation (especially in skeletal muscle).
 - High levels: α -receptor activation $\rightarrow$ vasoconstriction.
 - Noradrenaline $\rightarrow$ always a vasoconstrictor.
 - Other **vasoconstrictor hormones**: Angiotensin II and Vasopressin.

Intrinsic control

Extrinsic

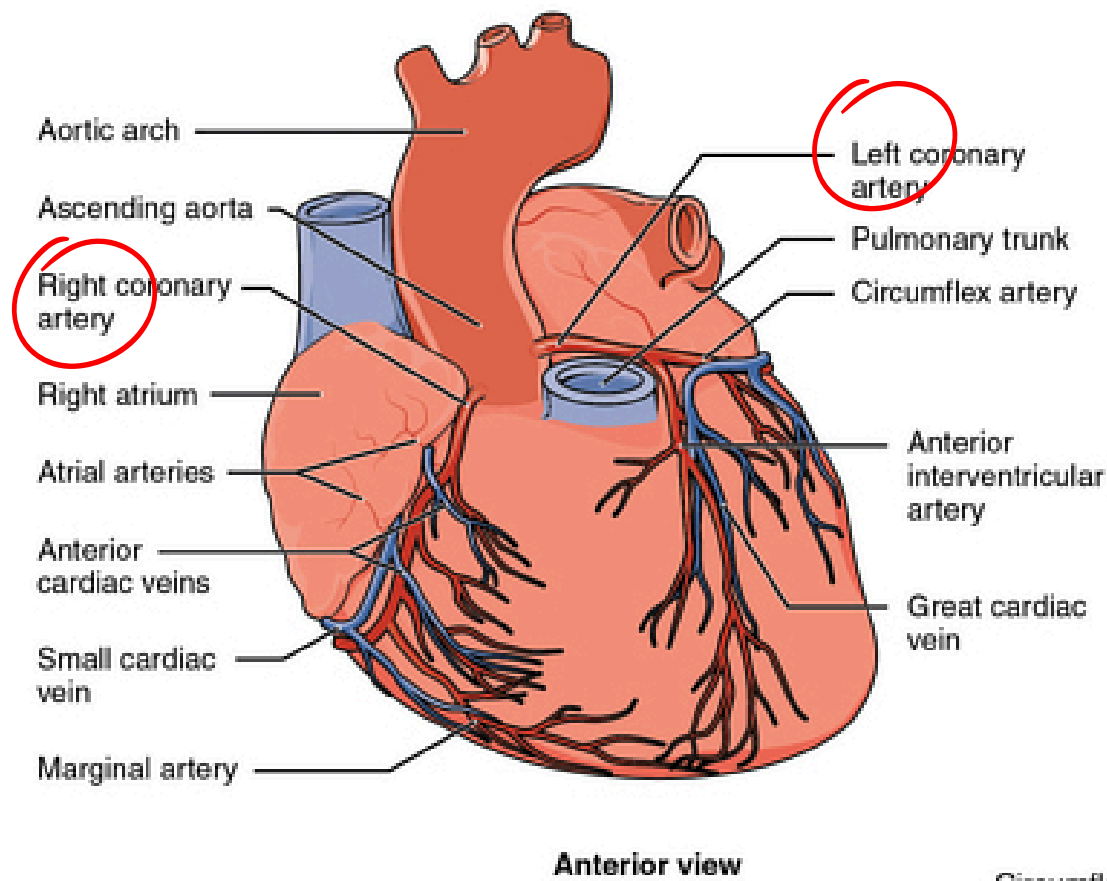
Intrinsic



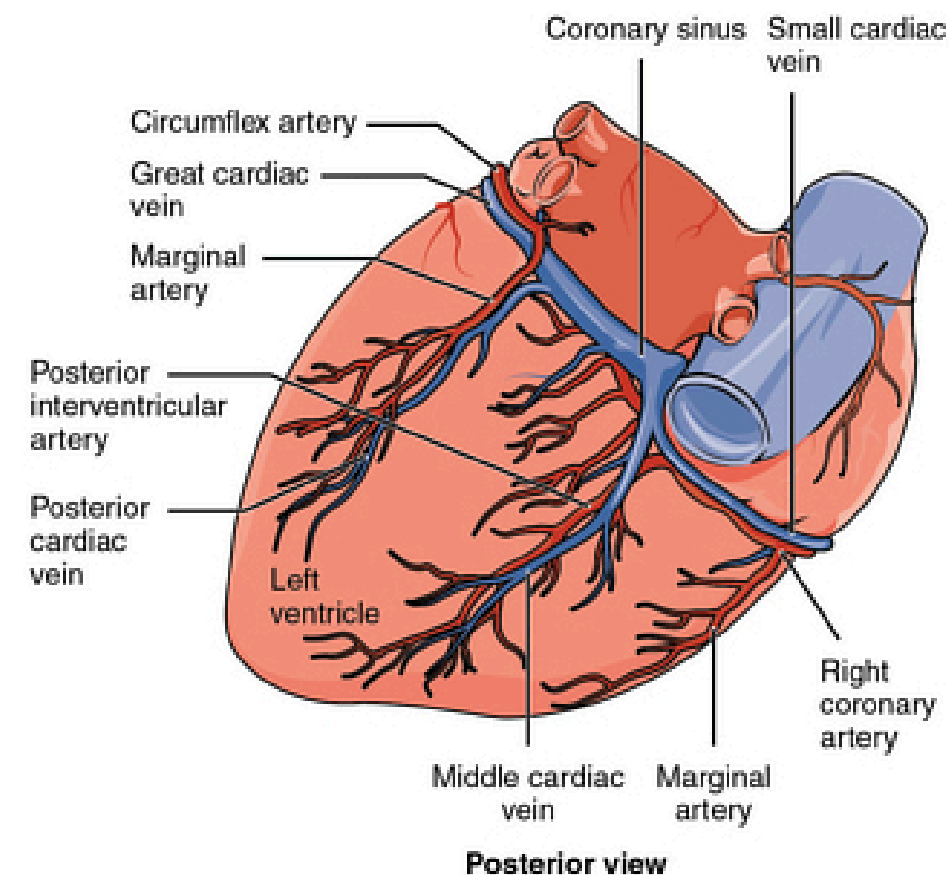
- Local control mechanisms act through:
 - **Direct effects on vascular smooth muscle**, and
 - Indirect effects mediated by the vascular endothelium.

Vasoconstrictor (+) and vasodilator (–) influences acting upon arterioles determine vascular tone. Extrinsic factors are from outside of the tissue, whereas intrinsic factors are from within the tissue or vessel. Tone is determined by the balance of all the factors.

Coronary circulation

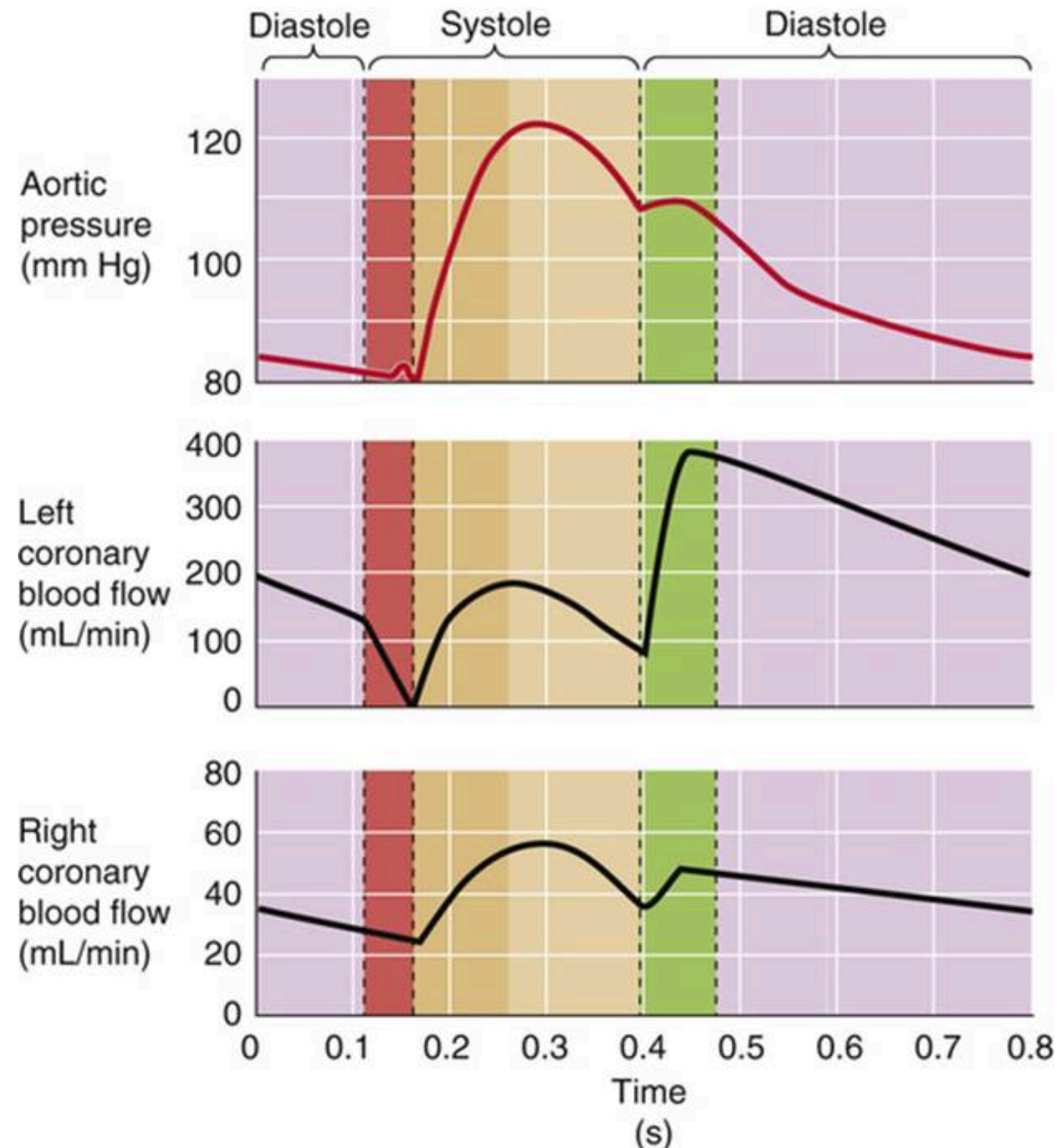


<https://radiopaedia.org/articles/coronary-arteries>

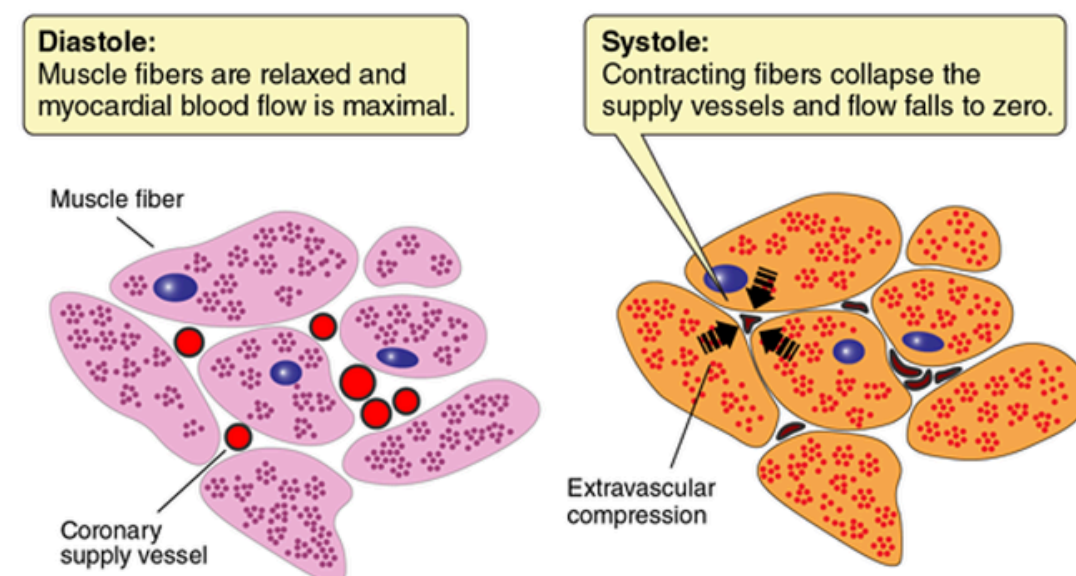


- The myocardium receives its entire blood supply from the coronary arteries.
 - **Right coronary artery (RCA):** supplies right atrium and ventricle.
 - **Left coronary artery (LCA):** divides into anterior descending and circumflex branches, supplying the left atrium and ventricle.

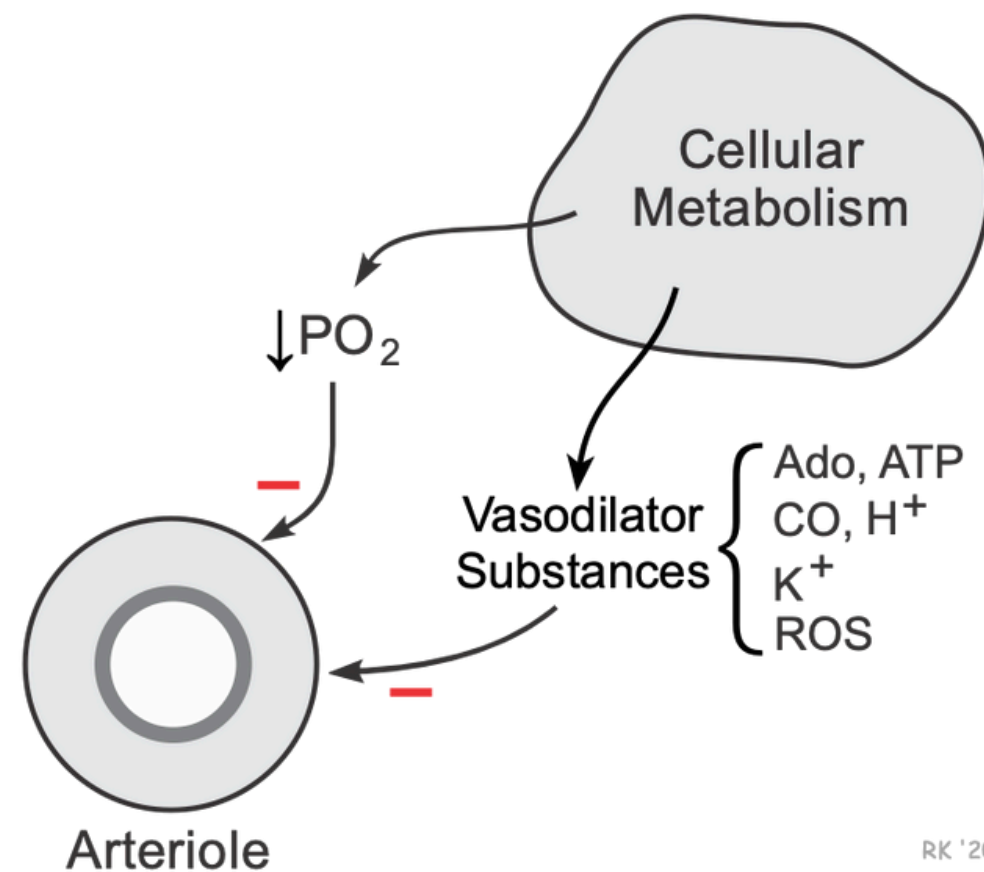
Control of Myocardial Blood Flow (CBF)



- Primary determinant: **Aortic pressure.**
- Unique feature: CBF is strongly influenced by mechanical activity of the heart.
 - **During systole: Coronary vessels are compressed,** reducing or reversing flow (especially in the left ventricle).
 - **During early diastole:** CBF is maximal, as extravascular compression decreases and aortic pressure remains high.
- Factors reducing CBF:
 - Tachycardia (shortened diastole)
 - High ventricular end-diastolic pressure
 - Low arterial pressure
- **Endocardial vessels (inner wall) are most prone to ischemia due to higher compression.**



Control of Myocardial Blood Flow (CBF)

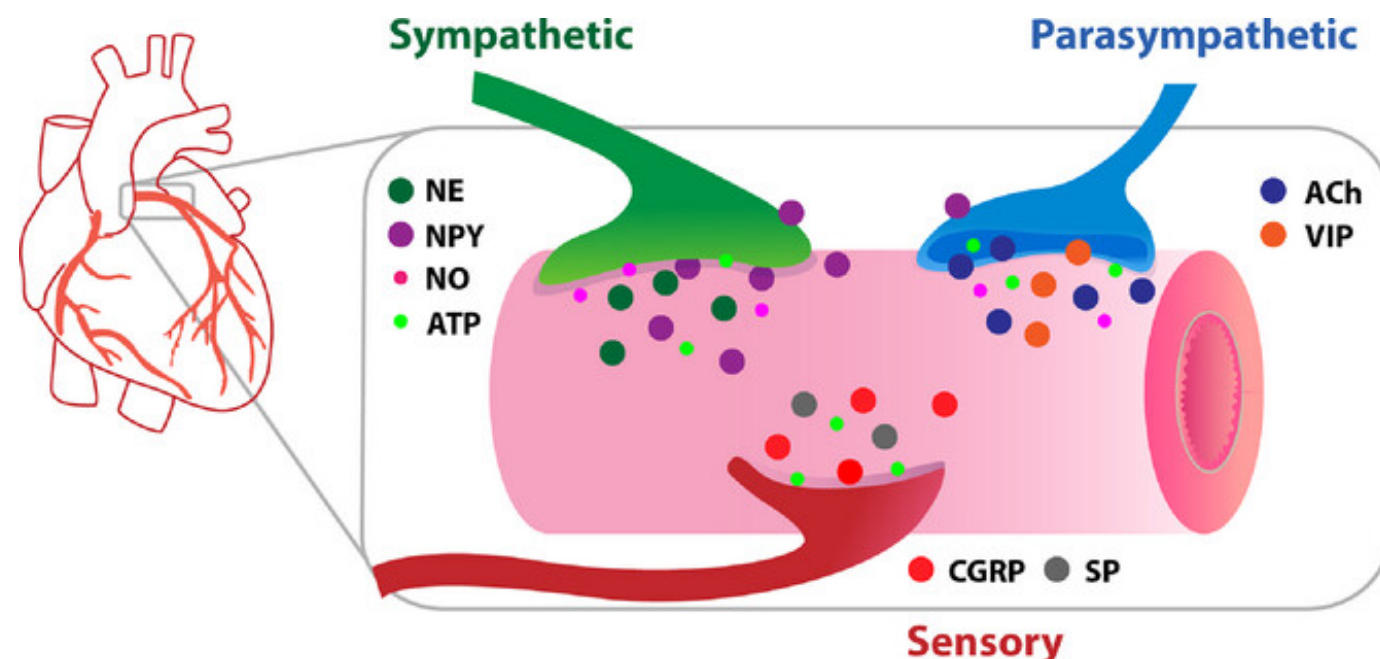


Metabolic Regulation

- CBF closely matches **myocardial oxygen consumption**.
- The heart extracts O₂ almost maximally under normal conditions, so increased demand requires increased flow.
- Adenosine contributes to metabolic vasodilation during ischemia, but less so during normal exercise.
- Other possible mediators: hypoxia, hypercapnia, K⁺, and H⁺ ions.

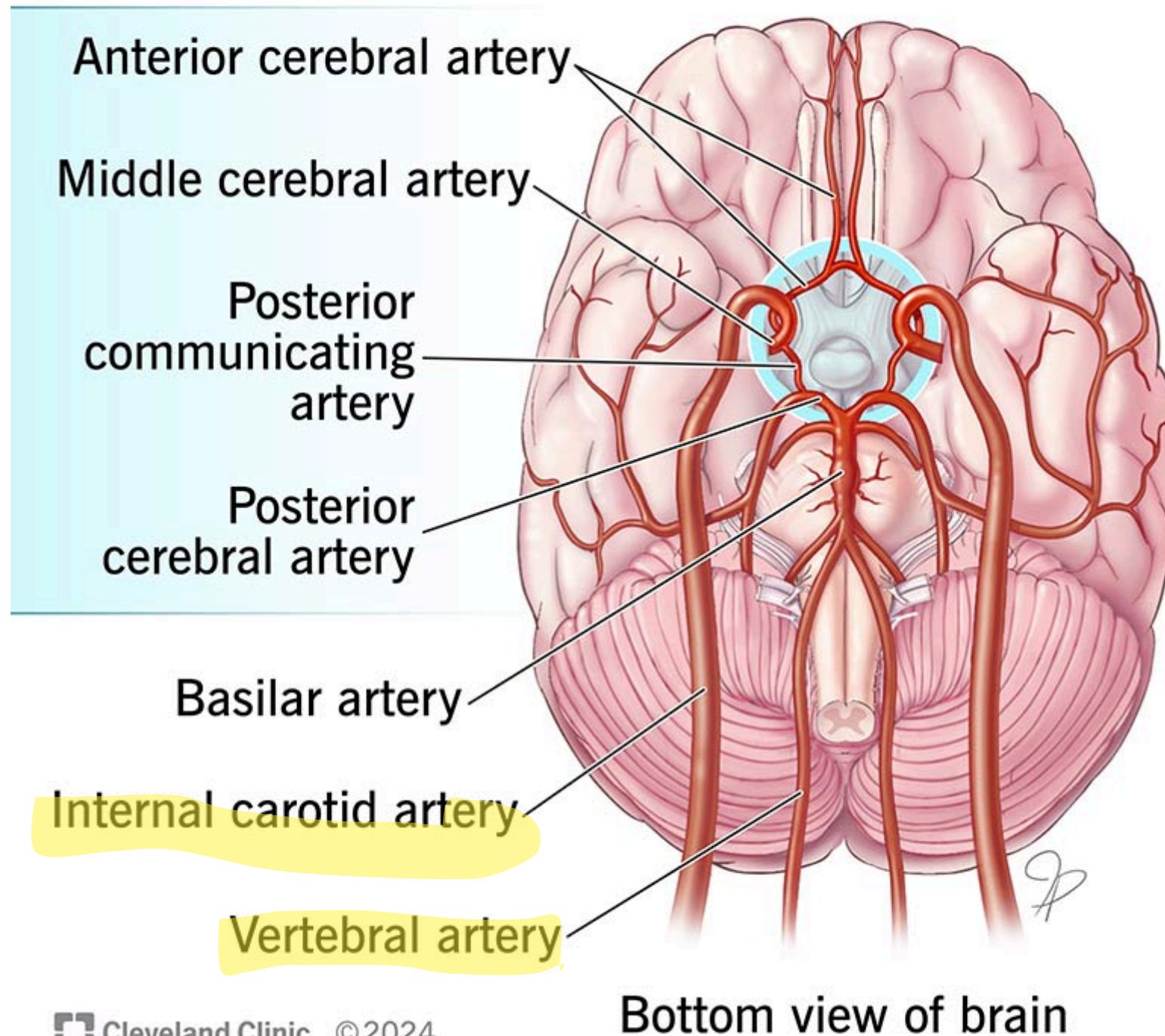
Neural Influence

- Sympathetic activation: transient vasoconstriction.
- Vagal activation: mild vasodilation.
- **However, metabolic factors dominate overall coronary blood flow control.**



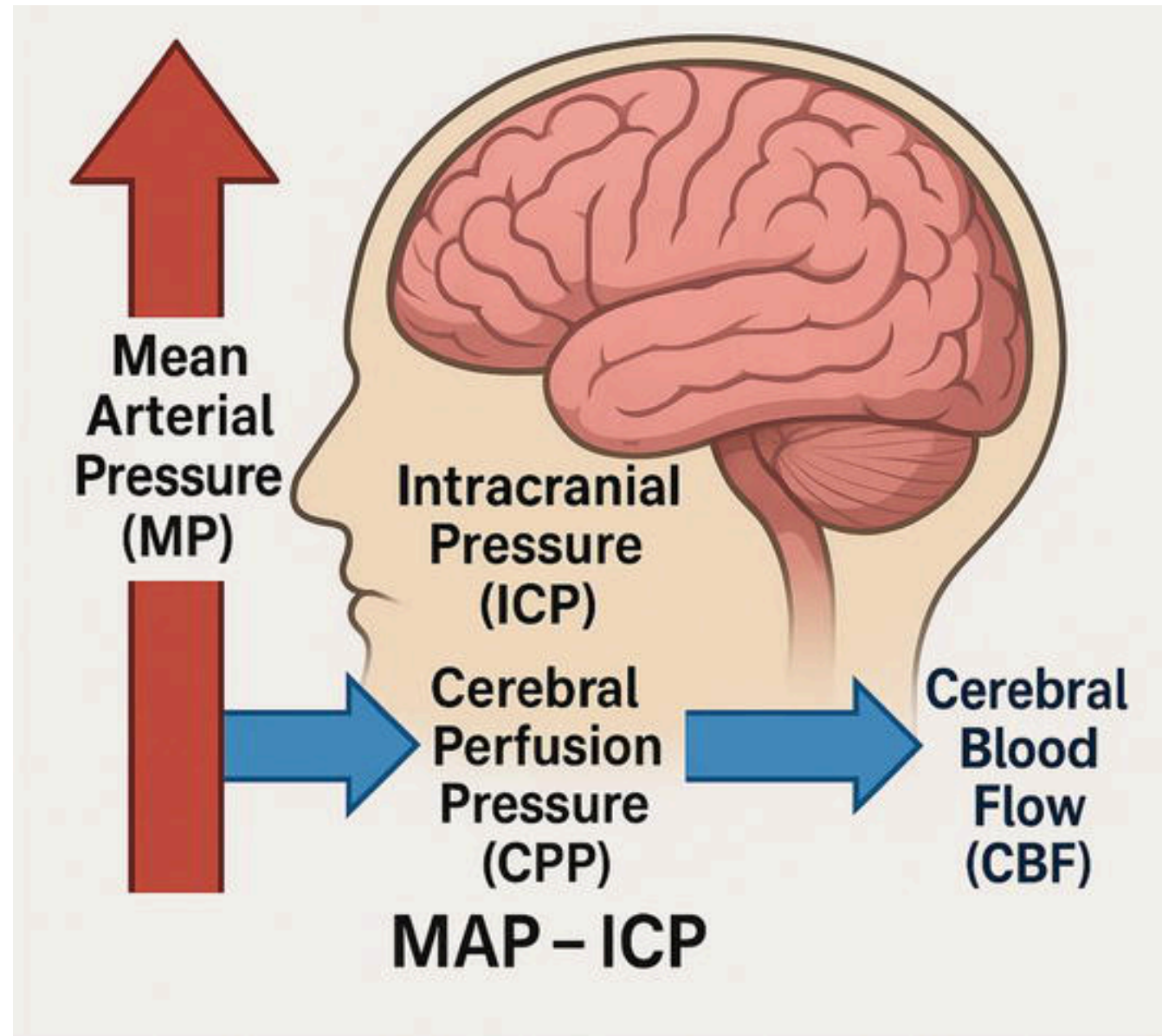
Cerebral circulation

Circle of Willis



- Blood supply: via **internal carotid and vertebral arteries**, which join to form the circle of Willis.
- The brain is highly sensitive to ischemia:
 - Blood flow interruption for ~5 seconds causes loss of consciousness.
 - Irreversible damage occurs within minutes.
- During reduced cardiac output (e.g., hemorrhage), cerebral blood flow (CBF) is prioritized over other organs.

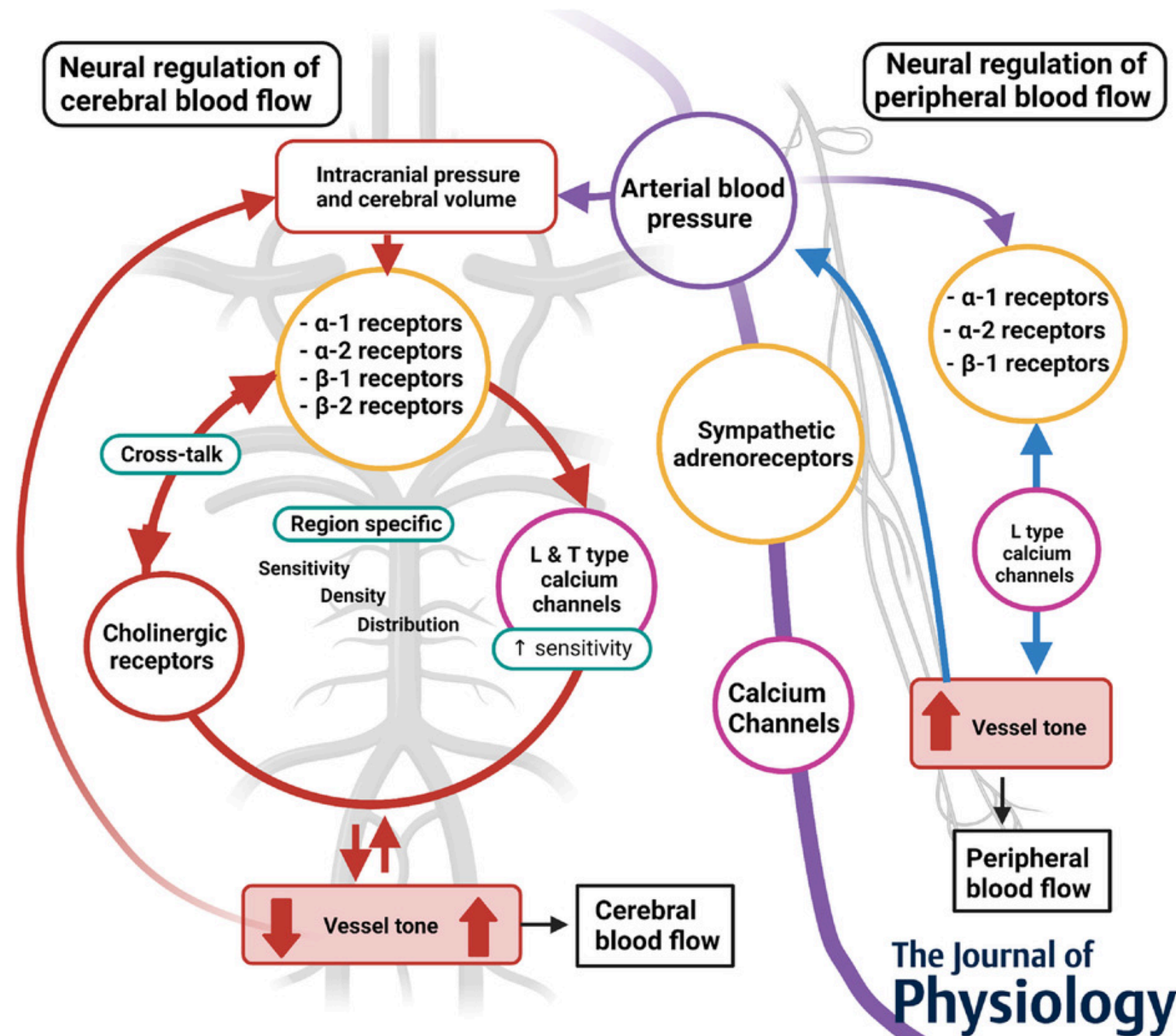
Intracranial Pressure and Perfusion



- The brain is enclosed within the rigid cranium; its contents are incompressible.
- Head injury may cause cerebral edema → increased **intracranial pressure (ICP)**.
- Under these conditions, **cerebral perfusion pressure (CPP) = arterial pressure – ICP**.
 - Elevated ICP reduces CBF, risking cerebral ischemia.
- Hypertonic resuscitation fluids may help lower ICP after head injury.

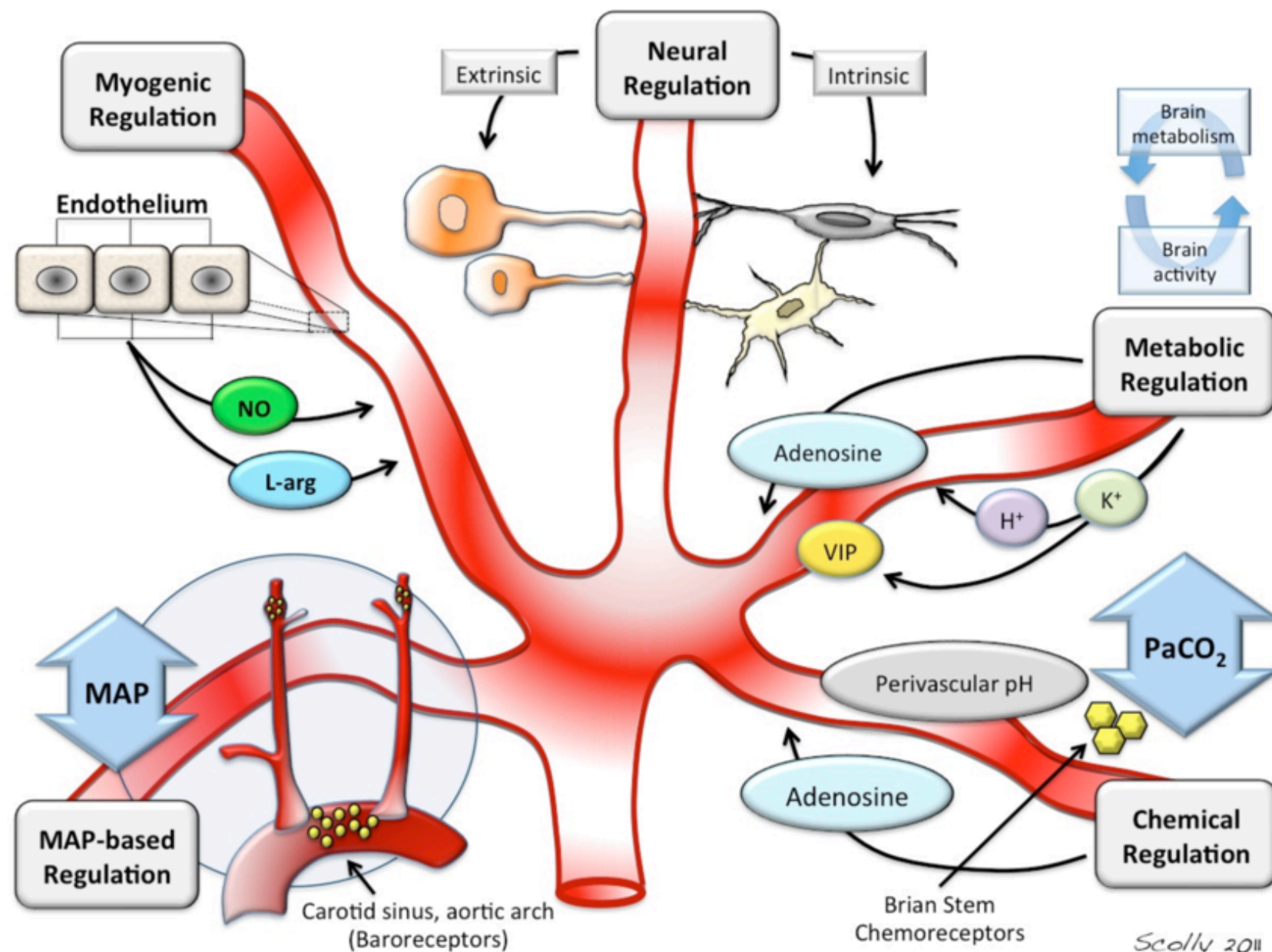
normal CPP: about 60–80 mmHg
if CPP < 50 mmHg → hypoperfusion (ischemic)

Control of Cerebral Blood Flow



- Total cerebral blood flow (CBF) is tightly regulated to maintain continuous oxygen delivery.
- Local CBF increases with neuronal activity via **metabolic mediators**.
- Extrinsic control (e.g., sympathetic) is weak, mainly influencing large cerebral arteries.
- **Intrinsic control dominates and regulates cerebral resistance vessels.**
 - **Autoregulation**
 - CBF remains constant between 60–180 mmHg arterial pressure (via **myogenic mechanism**).

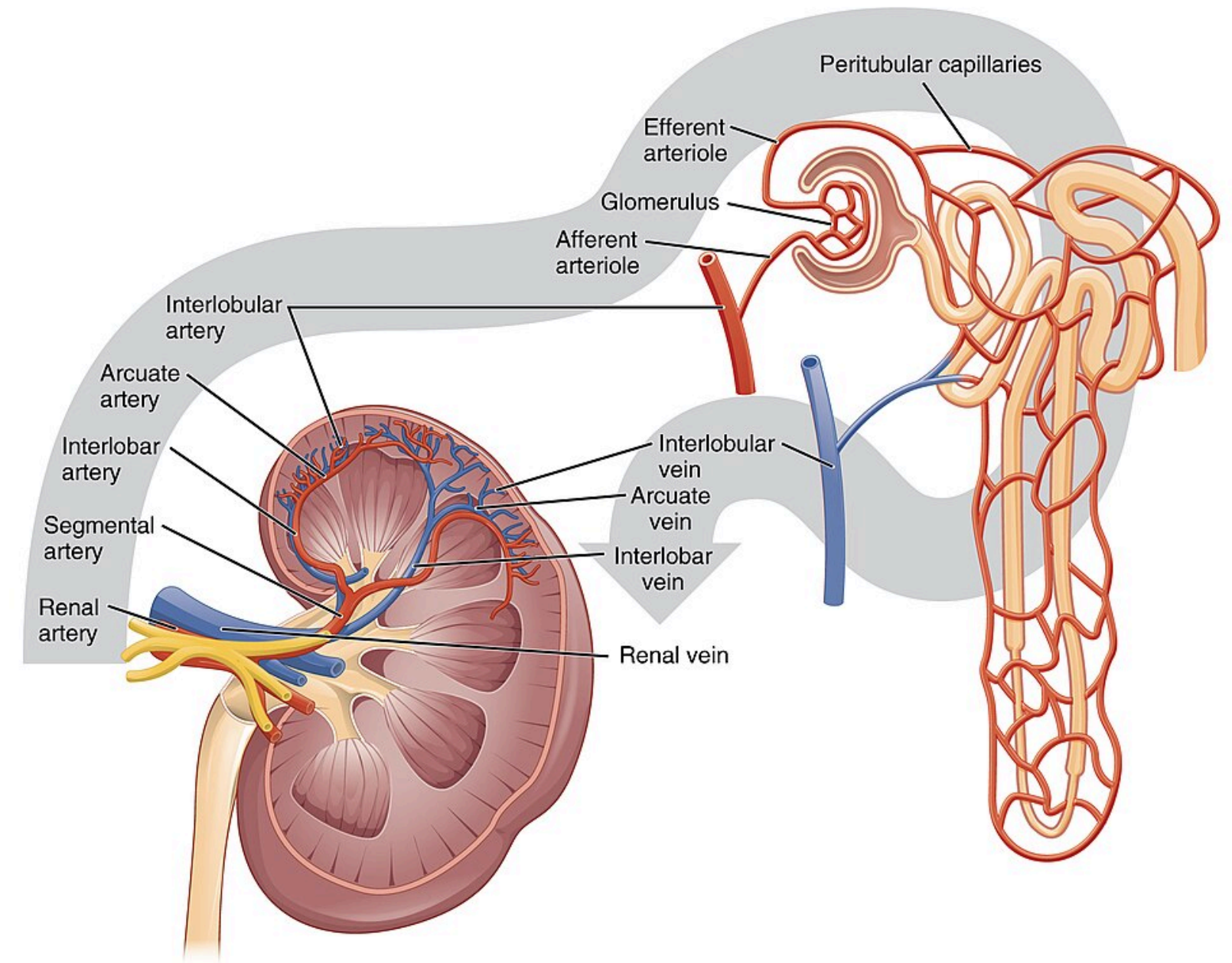
Chemical Regulation



- **Adenosine**
- **CO₂ tension (PaCO₂):**
 - Hypercapnia → vasodilation
 - Hypocapnia → vasoconstriction
- **O₂ tension (PaO₂):**
 - Hypoxemia (< ~8.5 kPa) → vasodilation
- **Nitric oxide (NO)** mediates vasodilation in response to hypercapnia, but not to hypoxemia.

Renal circulation

- Blood reaches the kidneys via **renal arteries**, which branch into **afferent arterioles → glomerular capillaries → efferent arterioles**.
- This unique arrangement allows precise control of both renal blood flow (RBF) and glomerular filtration rate (GFR).



Mechanisms Controlling RBF

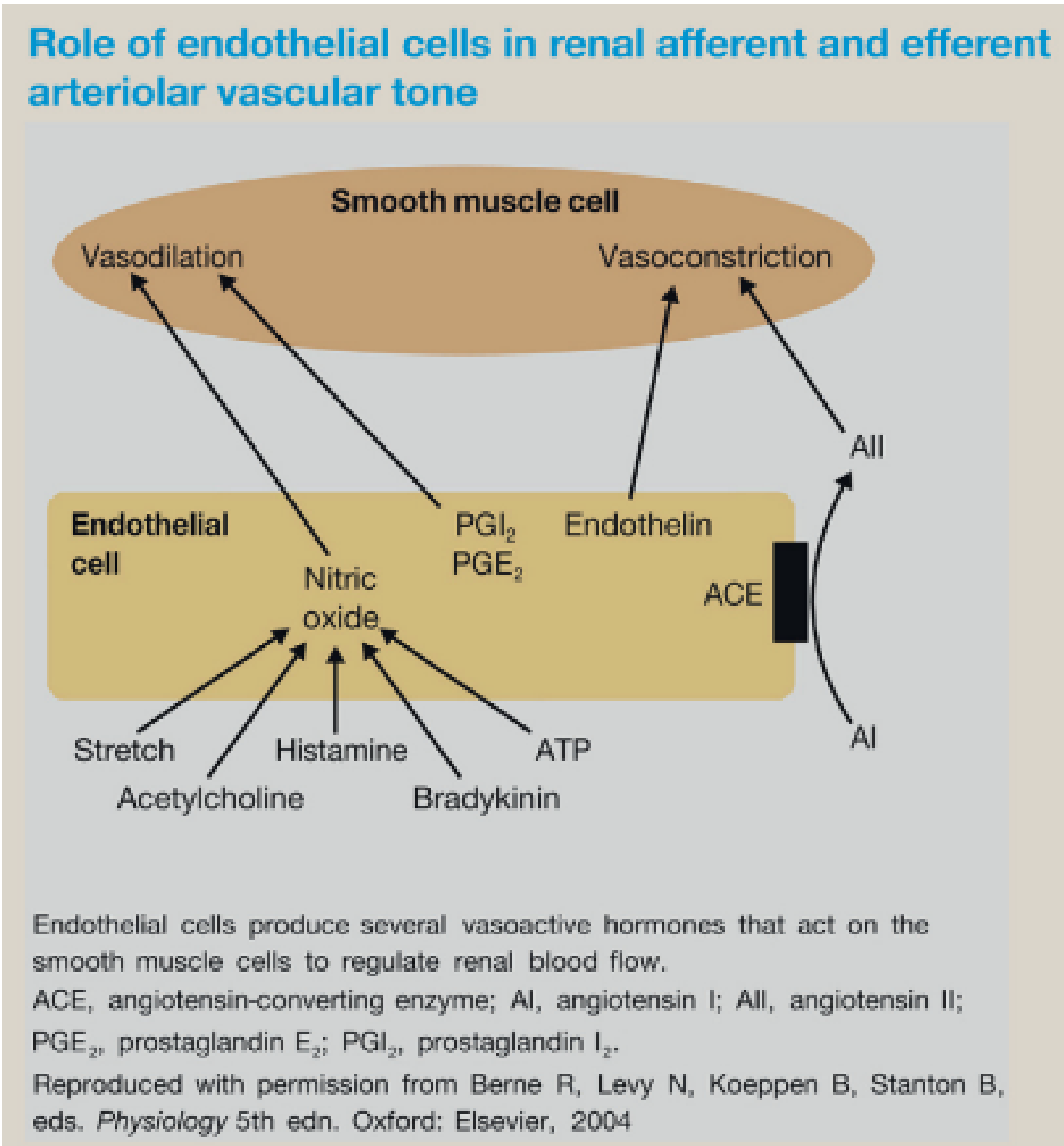
Mechanism	Contribution	Response Time	Key Features
Myogenic Mechanism (MM)	~50%	<10 sec	Vascular smooth muscle contracts when stretched → prevents excessive flow.
Tubuloglomerular Feedback (TGF)	35–50%	30–60 sec	Macula densa senses ↑NaCl in distal tubule → constricts afferent arteriole → ↓GFR.
Third mechanism	<15%	30–60 sec	Exact nature unknown; more important at very low pressures.

1. Intrinsic (Autoregulation)

- Autoregulation maintains stable RBF and GFR when mean arterial pressure (MAP) is between 90–180 mmHg.

2. Extrinsic (Neural & Humoral)

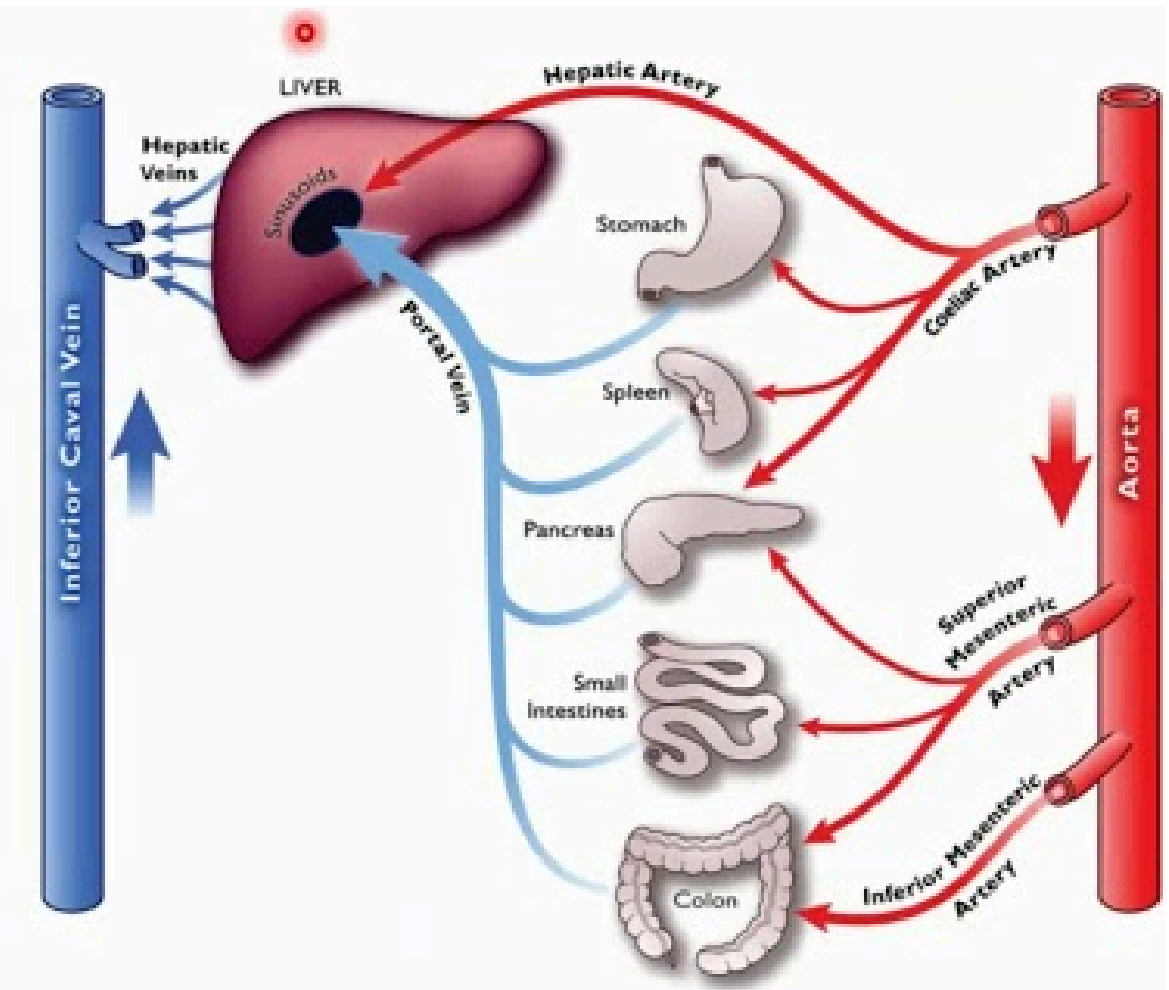
- Sympathetic nerves (α-adrenergic):
 - Cause vasoconstriction, mainly in afferent arterioles → ↓RBF & ↓GFR.
 - Triggered by stress, strong emotion, or low effective circulating volume.
- Endothelial factors:
 - Vasodilators: Nitric oxide (NO), prostaglandins.
 - Vasoconstrictors: Endothelin, angiotensin II.



Splanchnic Circulation

- The splanchnic circulation supplies blood to the gastrointestinal organs, spleen, pancreas, and liver.
- It receives approximately 25–30% of cardiac output at rest, making it one of the largest vascular beds in the body.
- The splanchnic system also serves as a **large blood reservoir**, containing up to 20% of total blood volume, which can be mobilized during stress or haemorrhage.

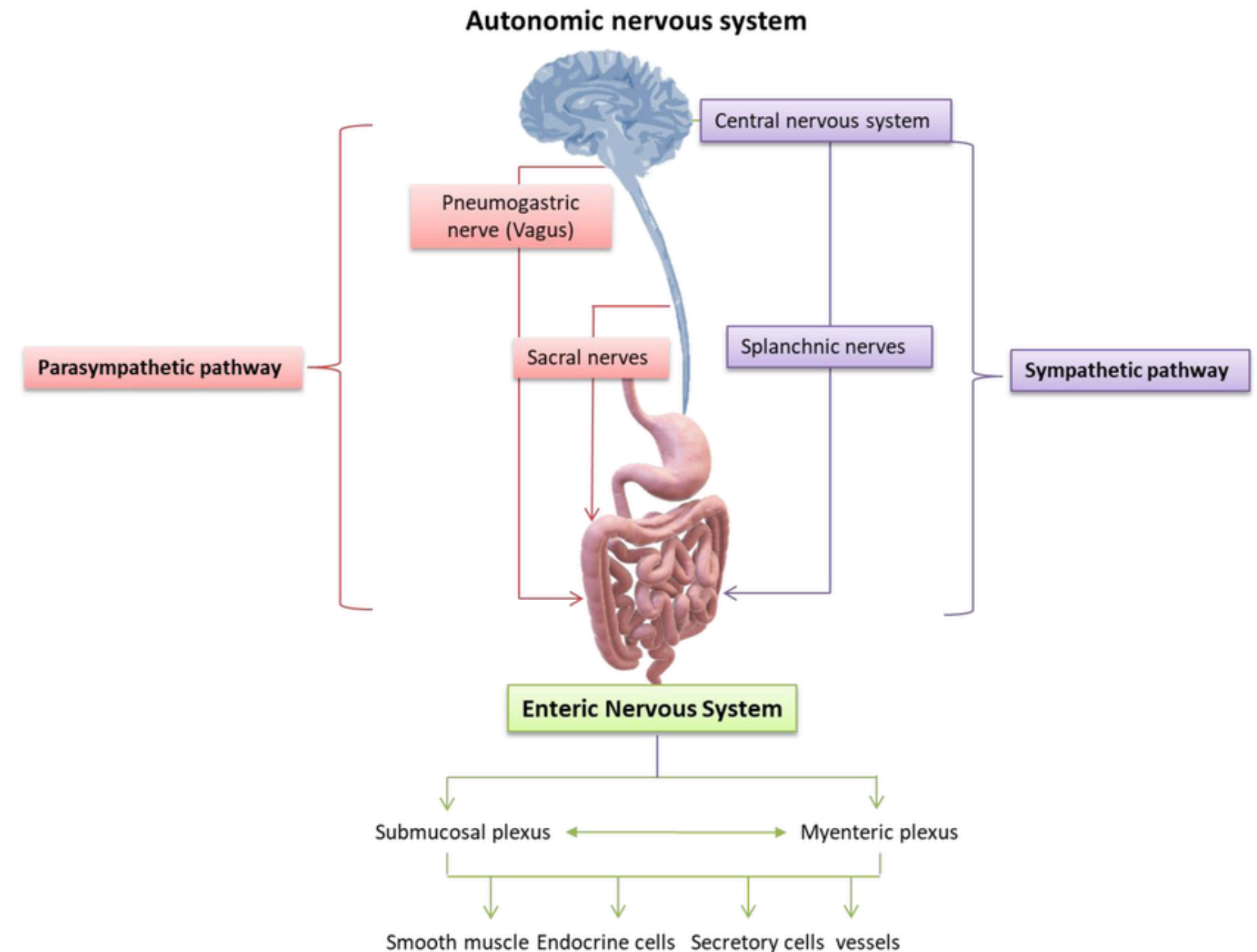
- Arteries that branch off aorta to serve digestive organs
 - Hepatic, splenic, and left gastric arteries
 - Inferior and superior mesenteric arteries
- **Hepatic portal circulation**
 - Drains nutrient-rich blood from digestive organs
 - Delivers blood to liver for processing



Control of Splanchnic Blood Flow

1. Extrinsic (Neural) Control

- Sympathetic adrenergic vasoconstrictor fibres regulate vascular tone through α_1 -adrenoceptors.
 - Increased sympathetic activity → vasoconstriction → reduced flow and blood pooling into central circulation.
 - Occurs during exercise, haemorrhage, or stress (e.g., “fight or flight” response).
- Parasympathetic stimulation (vagal) → mild vasodilation, mostly secondary to increased metabolic activity during digestion.



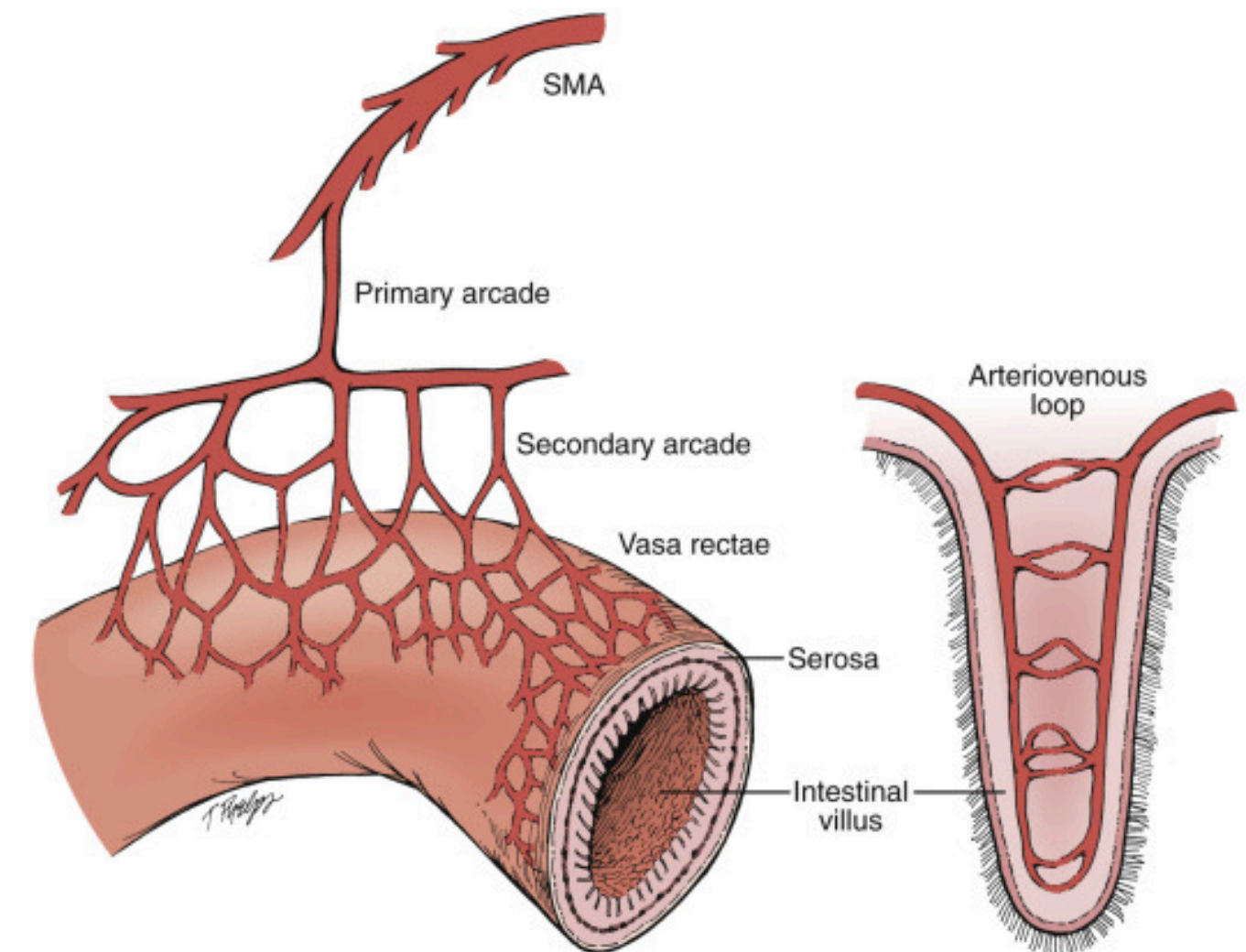
Control of Splanchnic Blood Flow

2. Intrinsic (Local) Control

- Metabolic vasodilation occurs in response to increased activity (digestion).
 - Mediators: adenosine, CO_2 , H^+ , K^+ , NO , and gut hormones (CCK, secretin, VIP).
- Myogenic autoregulation helps stabilize flow over a moderate pressure range.
- Endothelial control: NO and prostaglandins contribute to local vasodilation.

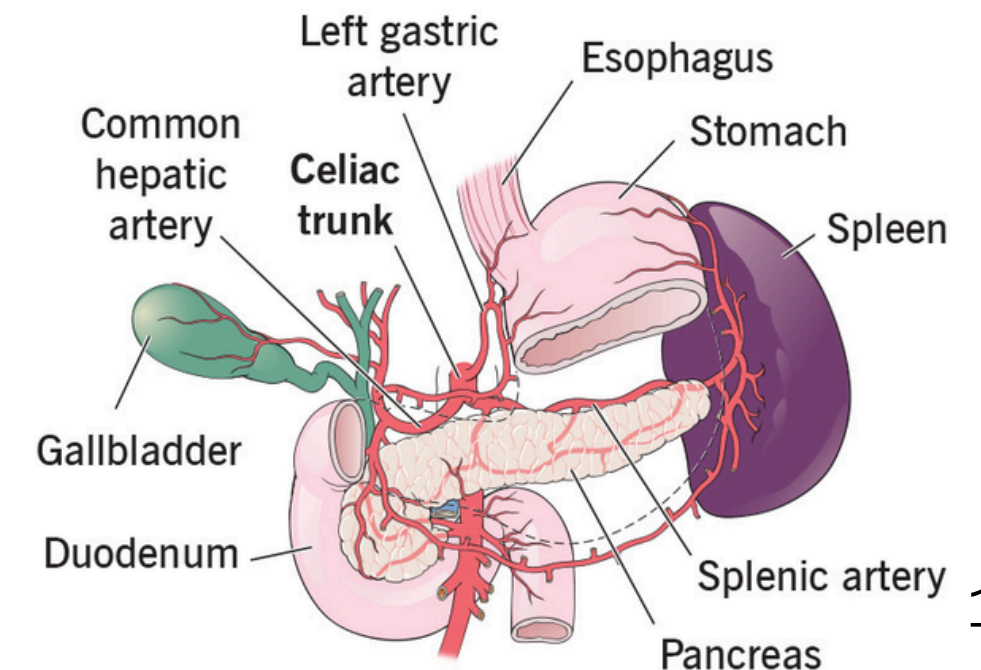
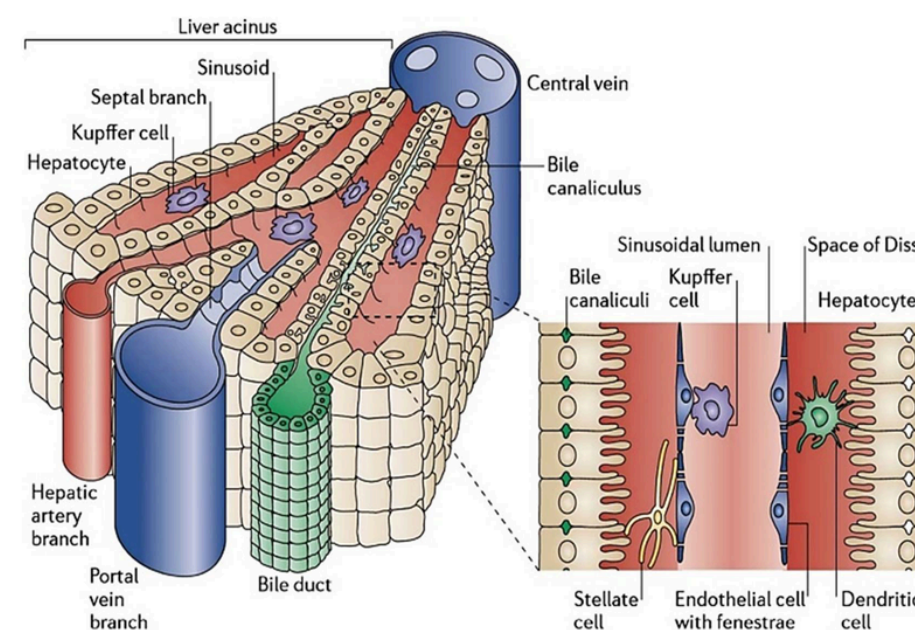
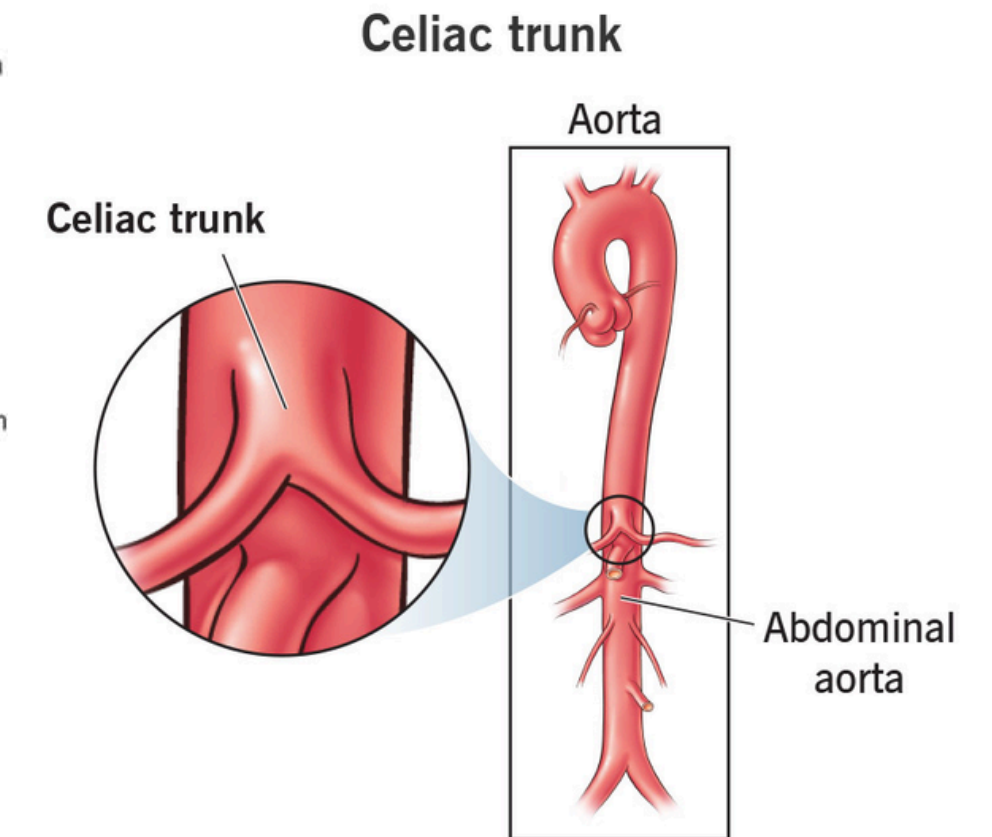
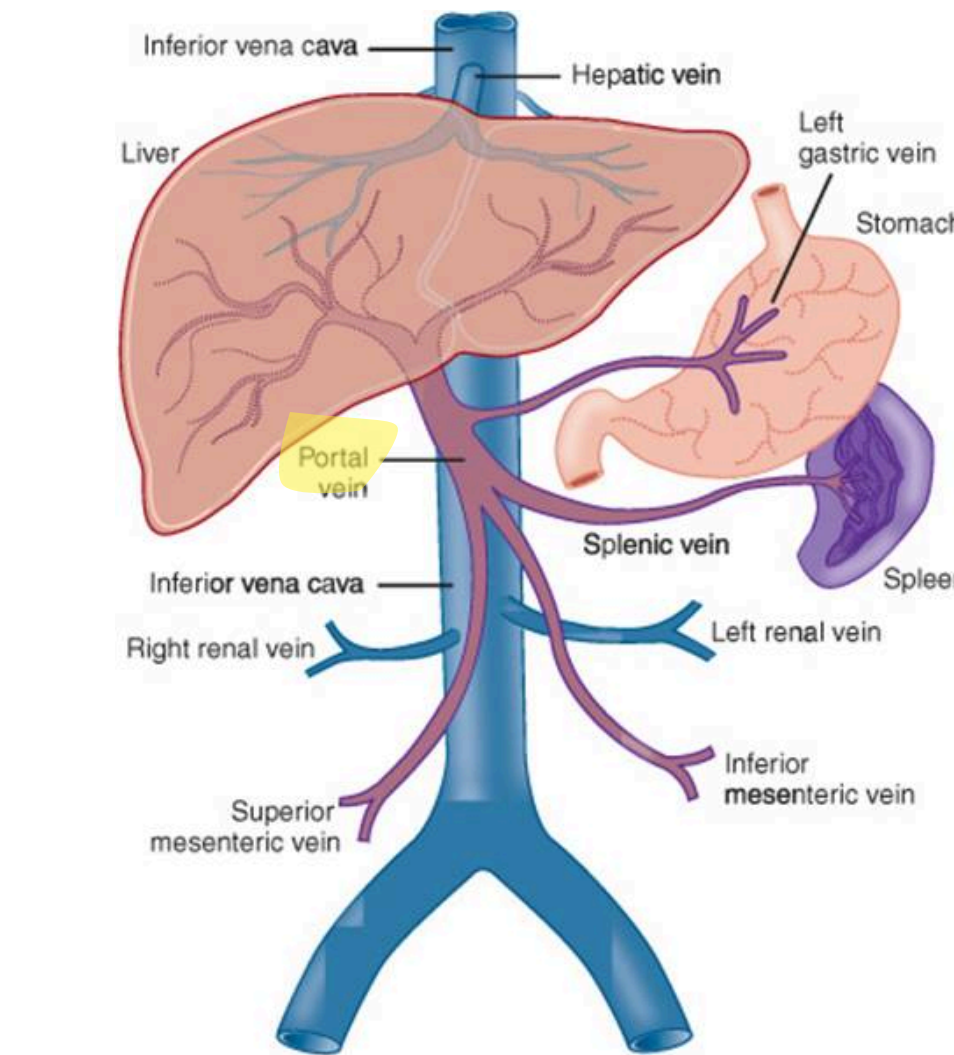
3. Humoral Factors

- **Vasodilators: gastrin, secretin, VIP, glucagon, cholecystikin (CCK).**
- Vasoconstrictors: angiotensin II, vasopressin, endothelin.
- These ensure appropriate redistribution of flow during digestion, fasting, or hypovolaemia.



Hepatic circulation

- The liver receives a dual blood supply:
 - **Portal vein (PV):** ~75% of total blood flow
 - Formed from the superior mesenteric vein and splenic vein
 - Carries partially deoxygenated blood from spleen & GI tract
 - **Hepatic artery (HA):** ~25% of total blood flow
 - Branch of the **coeliac trunk**
 - Carries fully oxygenated blood
 - Both PV and HA branch into portal venules and hepatic arterioles, which drain into sinusoids within the hepatic acinus.



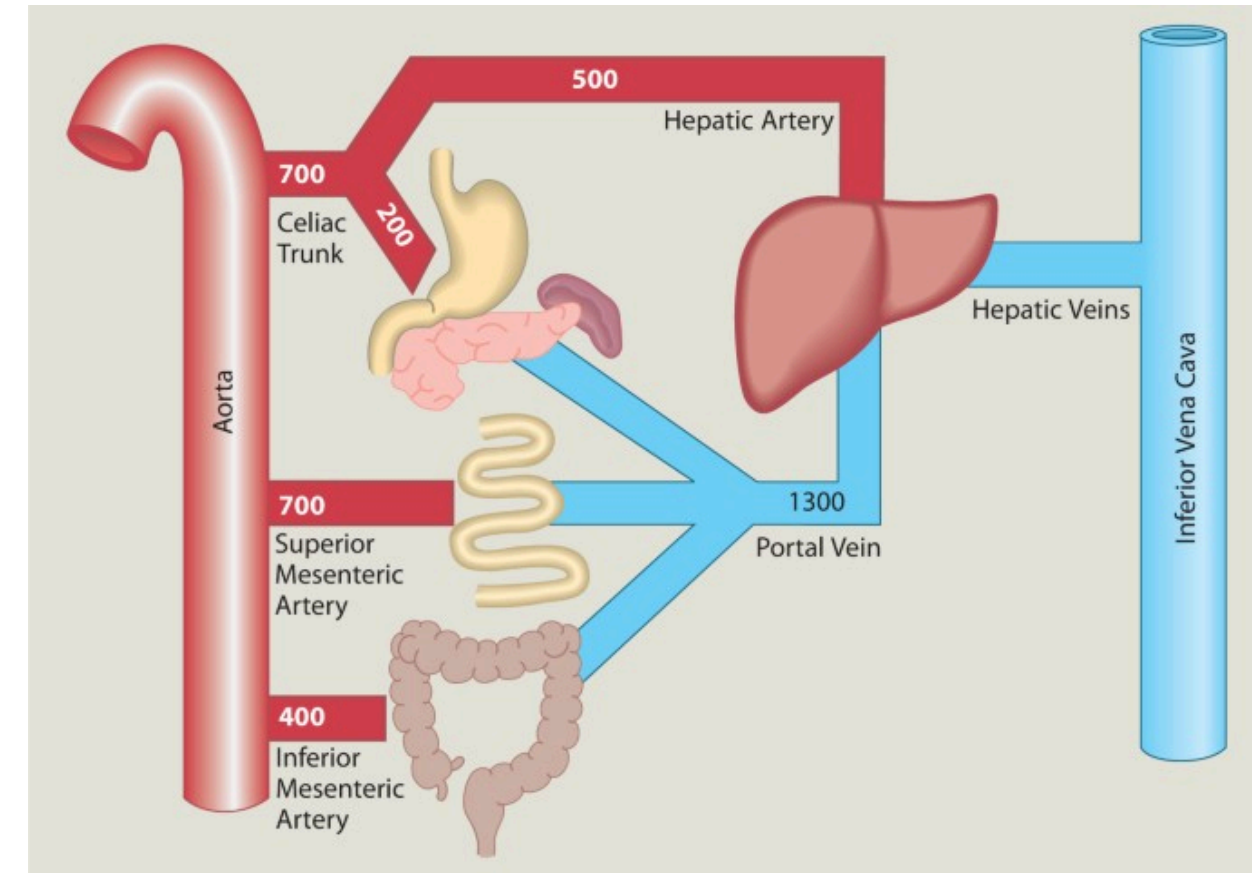
Control of Hepatic Blood Flow

1. Reciprocal Flow Regulation

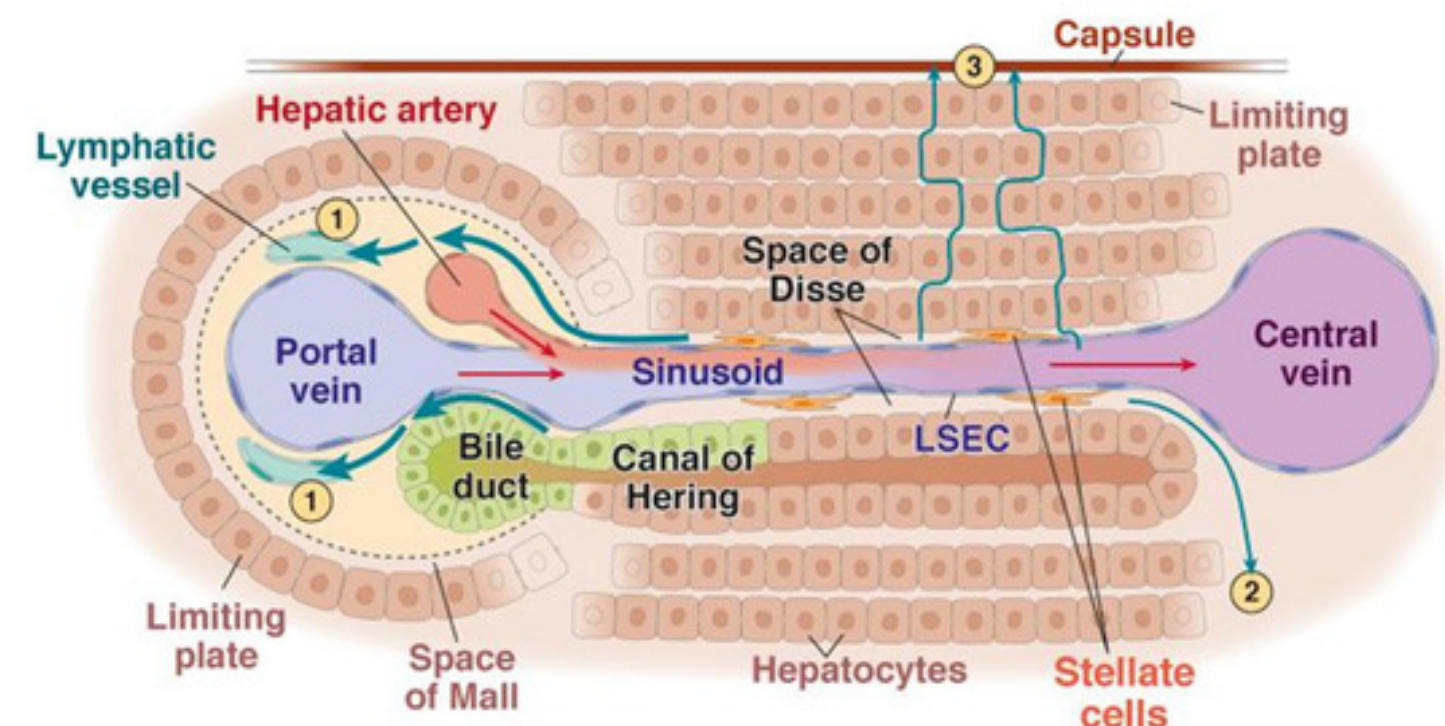
- Portal venous and hepatic arterial flows vary **inversely** to help maintain total hepatic blood flow.
- However, compensation is incomplete — one vessel cannot fully offset changes in the other.

2. Hepatic Arterial Buffer Response (HABR)

- Purpose: Maintain constant oxygen delivery when portal venous flow decreases.
- Mechanism:
 - Normal condition: Portal flow washes out adenosine.
 - ↓ Portal flow: **Adenosine accumulates**, causing hepatic arteriolar dilation → ↑ HA flow.
- Mediator: Adenosine (local vasodilator).
- Effect: Helps stabilize hepatic oxygen supply despite fluctuations in portal flow.



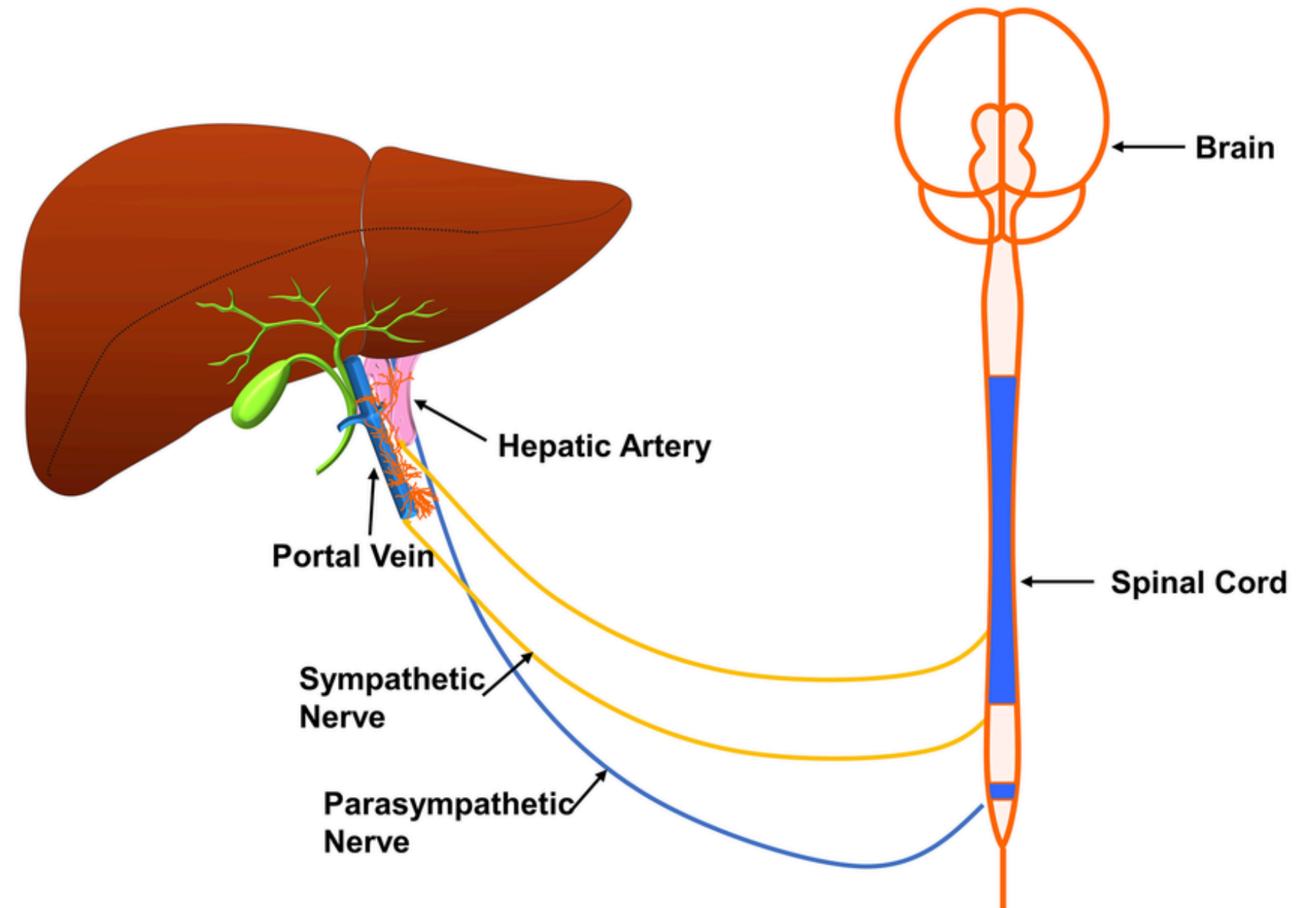
Lautt, W. W. (2010). Hepatic circulation: physiology and pathophysiology.



Control of Hepatic Blood Flow

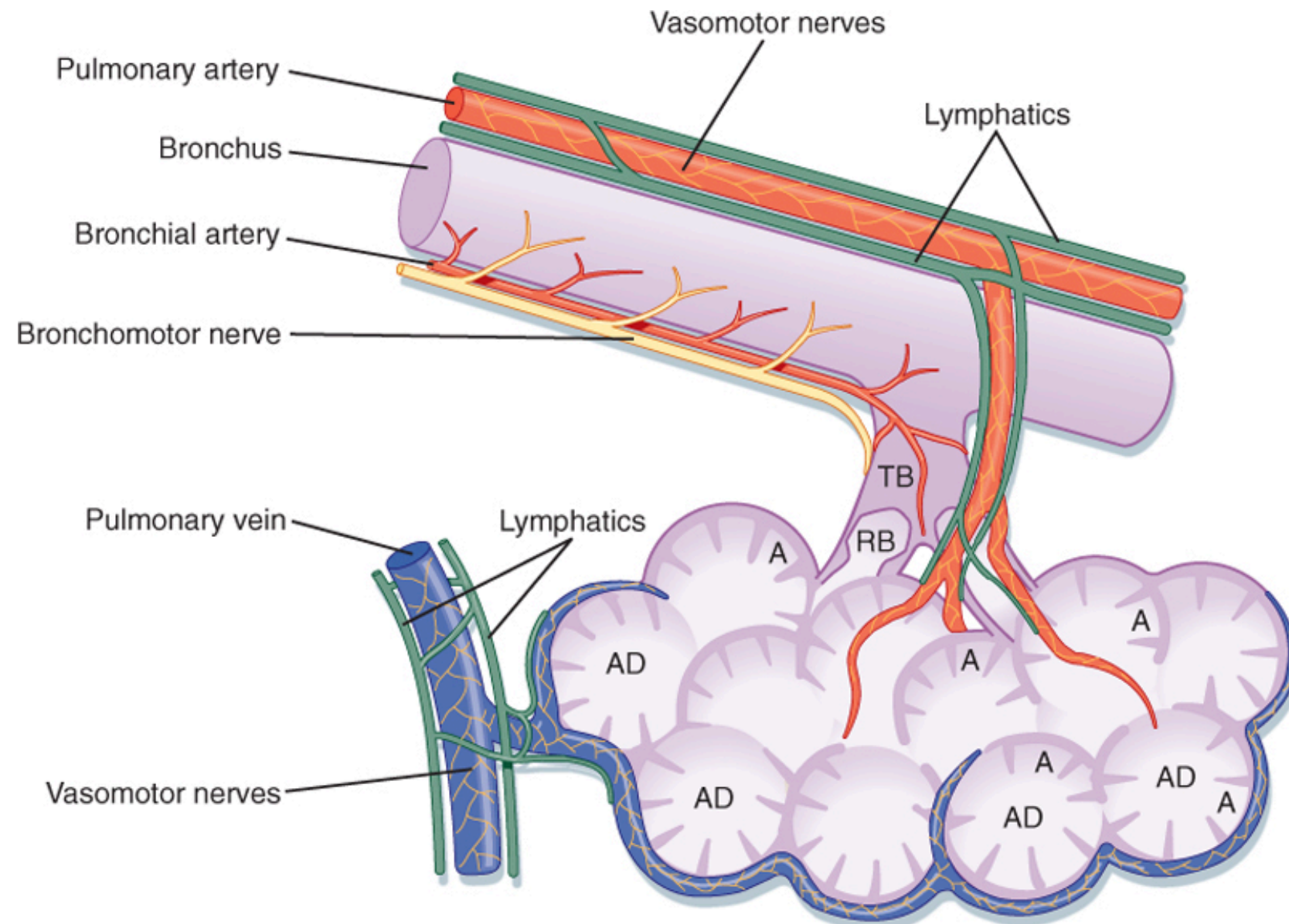
3. Neural Control

- Innervation: Both PV and HA receive sympathetic (noradrenergic) fibers.
- Effects:
 - Vasoconstriction → ↓ blood flow during hypovolemia or stress.
 - Capacitance effect: **Hepatic veins & sinusoids act as a blood reservoir** — sympathetic stimulation can mobilize ~50% of hepatic blood volume after hemorrhage.



Yang, K. et al., (2023).

Pulmonary circulation



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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Figure 20-8 Anatomic relationship between the pulmonary artery, the bronchial artery, the airways, and the lymphatics. A, alveoli; AD, alveolar ducts; RB, respiratory bronchioles; TB, terminal bronchioles.

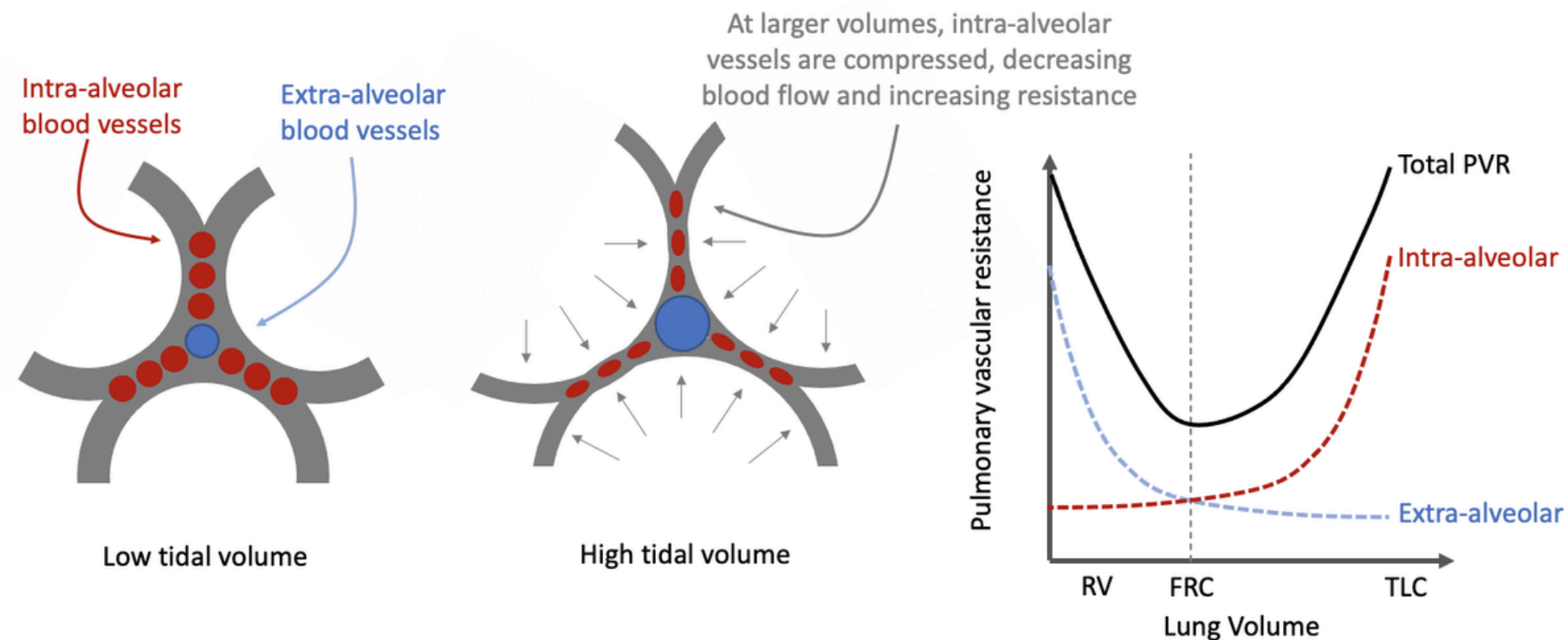
The pulmonary circulation is characterized as:

- Low resistance ($\approx$ one-tenth that of the systemic circulation)
- Low pressure
- High flow
- Extensive capillary network

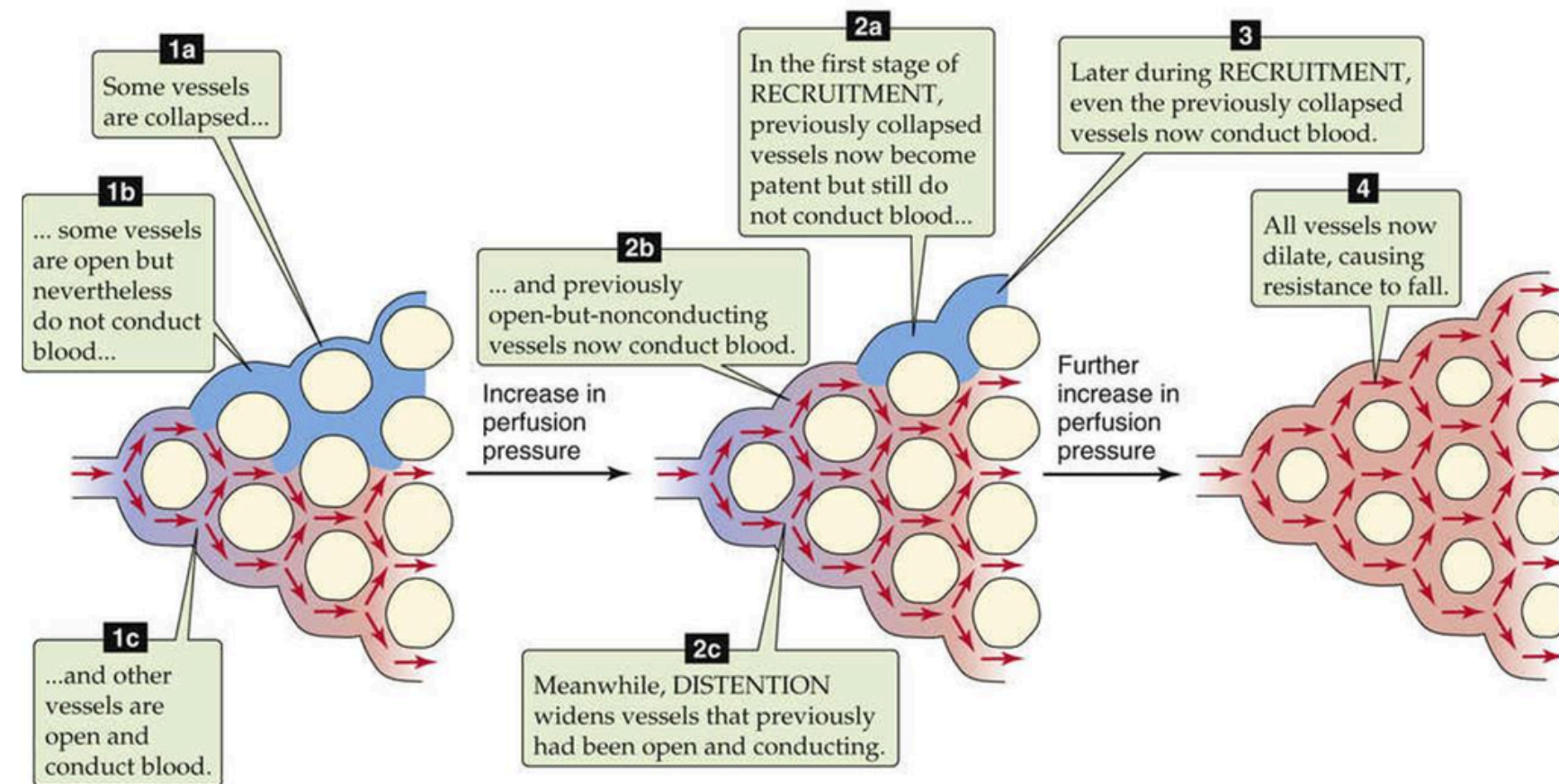
Pulmonary arterioles have minimal smooth muscle, limiting their ability to constrict like systemic arterioles.

- This design allows the lungs to accommodate the entire cardiac output continuously while minimizing the workload of the right ventricle.

Passive Regulation of Pulmonary Blood Flow

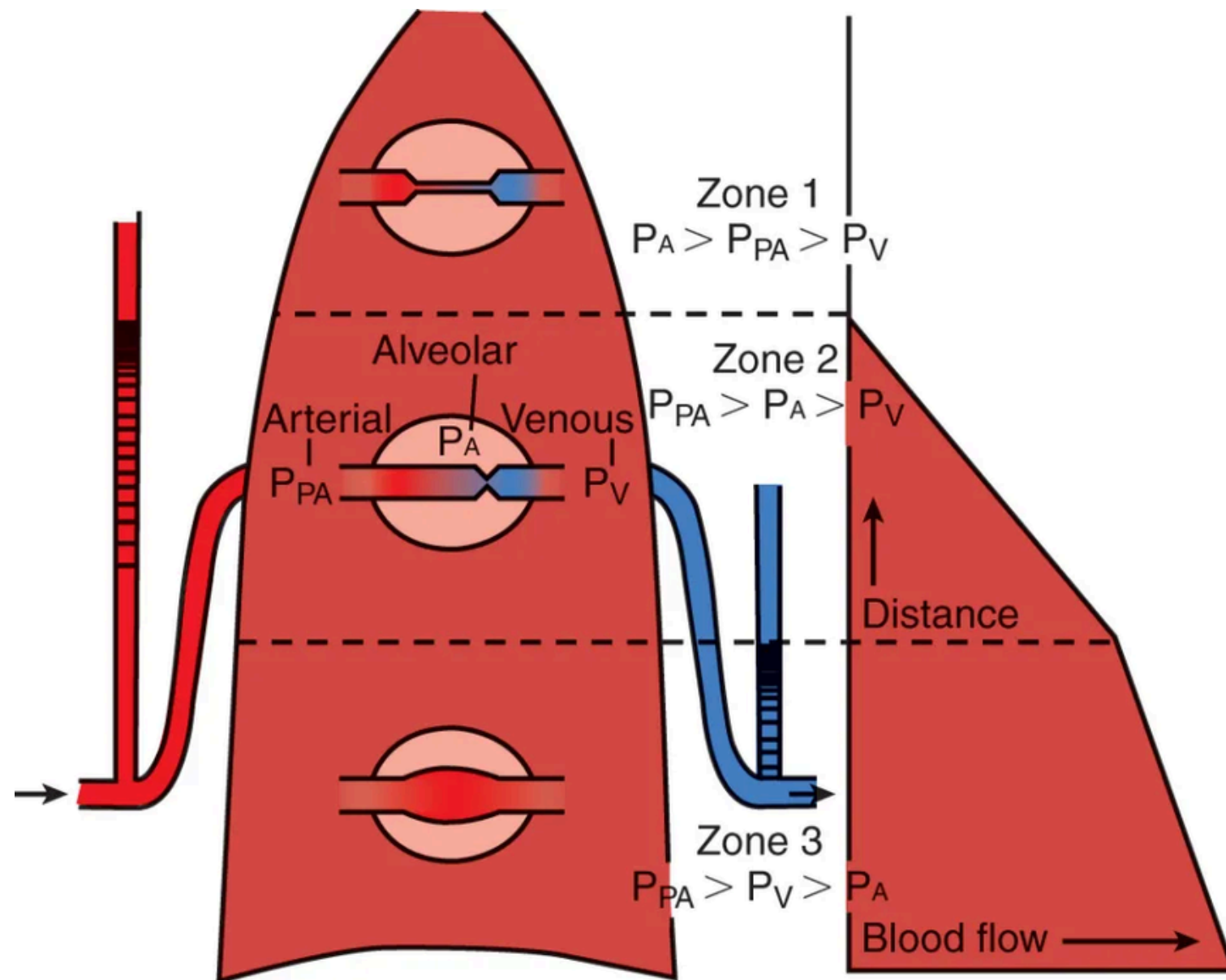


C RECRUITMENT AND DISTENTION OF ALVEOLAR VESSELS



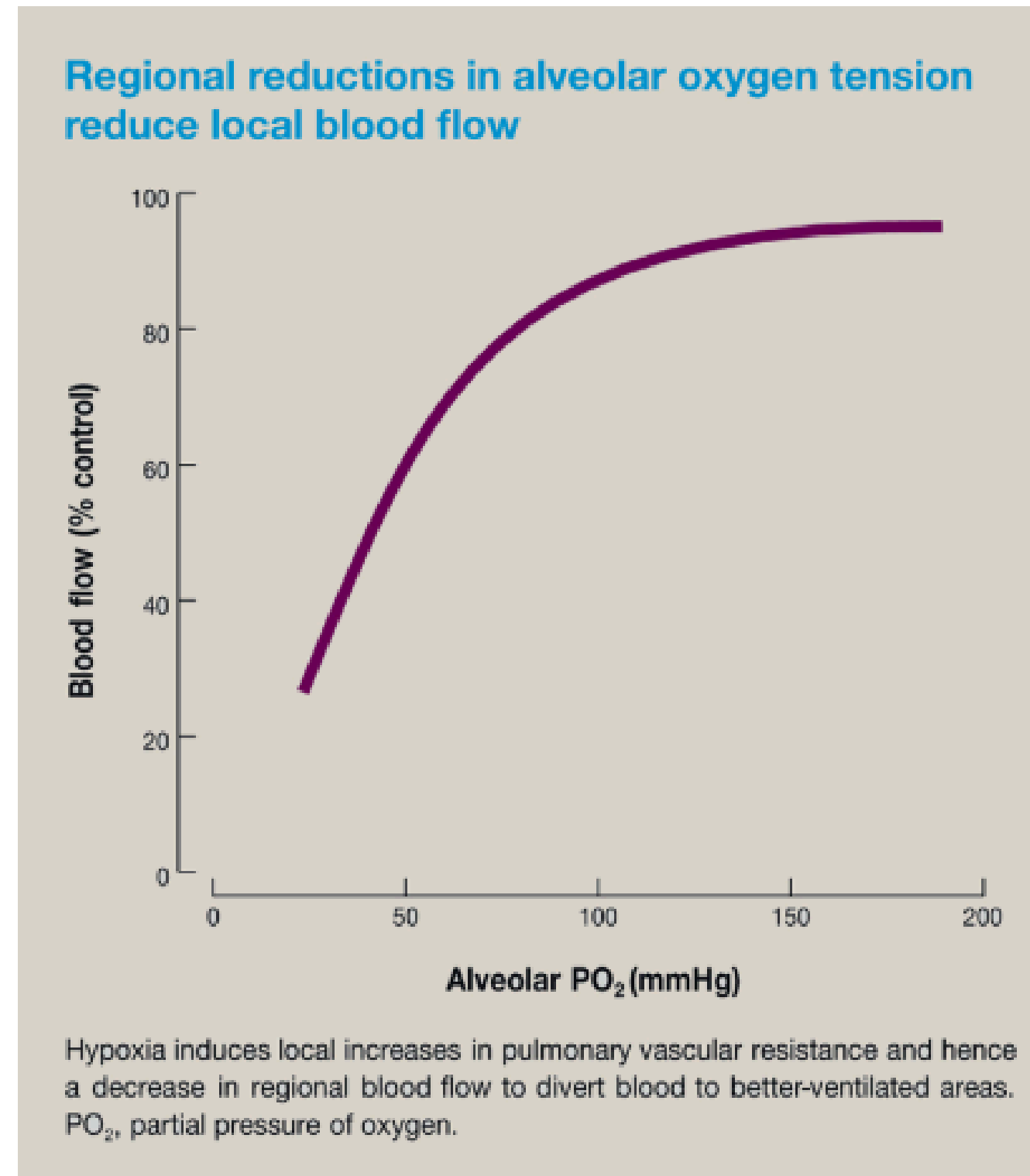
- Large increases in flow (e.g., during exercise) cause only small rises in pulmonary arterial pressure.
 - This is achieved by:
 - **Recruitment** (opening previously unperfused capillaries)
 - Distension (widening of existing vessels)
- **Lung volume effect on Pulmonary Vascular Resistance (PVR):**
 - PVR is highest at both low and high lung volumes.
 - PVR is lowest at Functional Residual Capacity (FRC).

Passive Regulation of Pulmonary Blood Flow



- Gravity and posture create regional variations in perfusion:
 - Upright position:
 - Highest flow: lung bases (due to hydrostatic pressure)
 - Lowest flow: lung apices
 - In hypovolemia or positive pressure ventilation, apical pressure may drop below alveolar pressure, leading to:
 - Cessation of local blood flow
 - Reduced gas exchange

Active Regulation of Pulmonary Blood Flow



Neural Control

- Pulmonary arterioles are innervated by the autonomic nervous system.
- Baroreceptor and chemoreceptor reflexes can modify PVR, though their role is relatively minor.

Hypoxic Pulmonary Vasoconstriction (HPV)

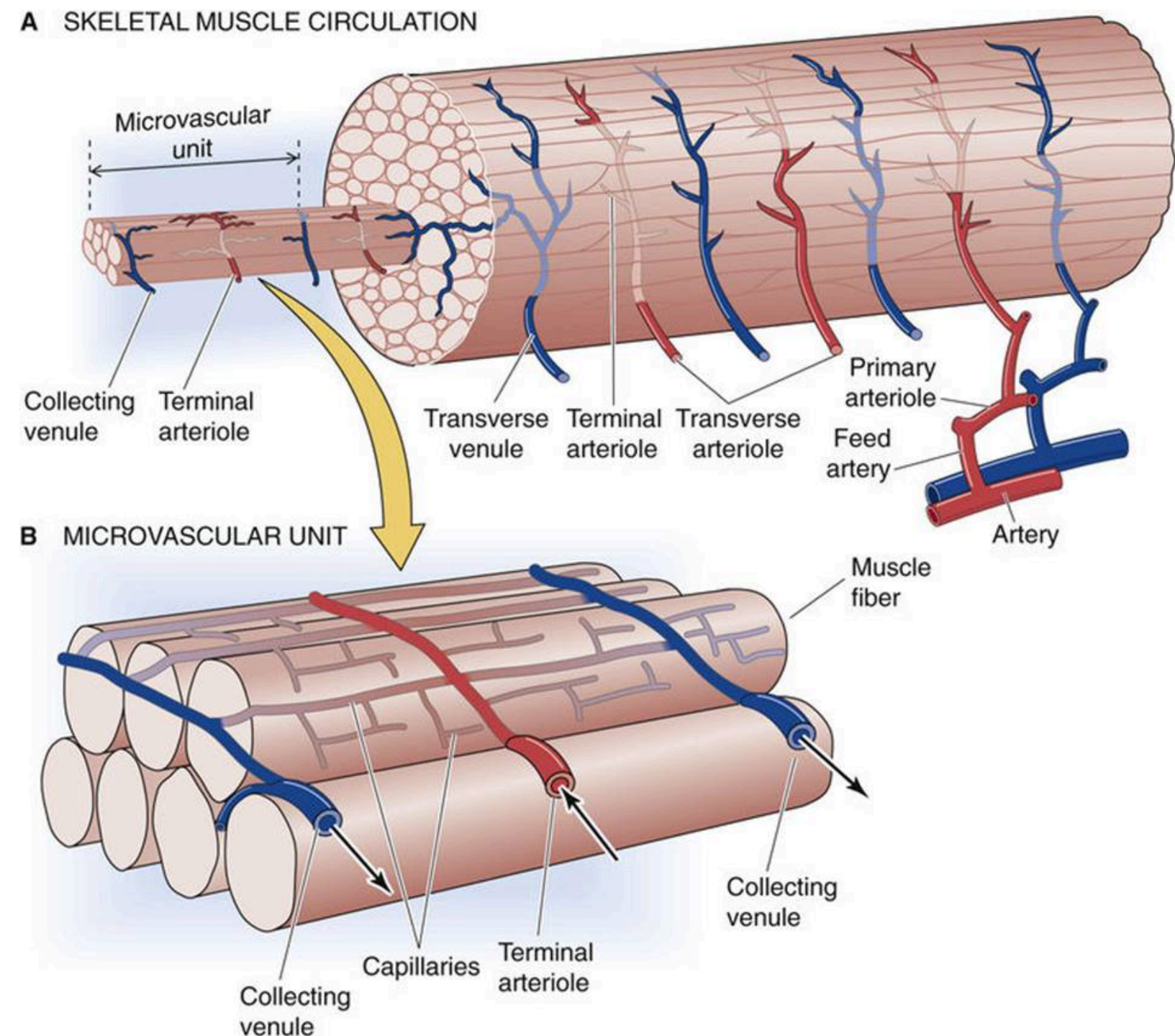
- Main mechanism of active control.
- Mechanism:
 - ↓ Alveolar O₂ tension → local arteriolar constriction
 - This response redirects blood toward better-ventilated alveoli, optimizing the ventilation–perfusion (V/Q) ratio.
- Clinical significance:
 - Local hypoxia → beneficial redistribution of flow.
 - Widespread hypoxia (>20% of lung) → generalized vasoconstriction → ↑ PVR, leading to:
 - Pulmonary hypertension
 - Pulmonary edema

Figure 3

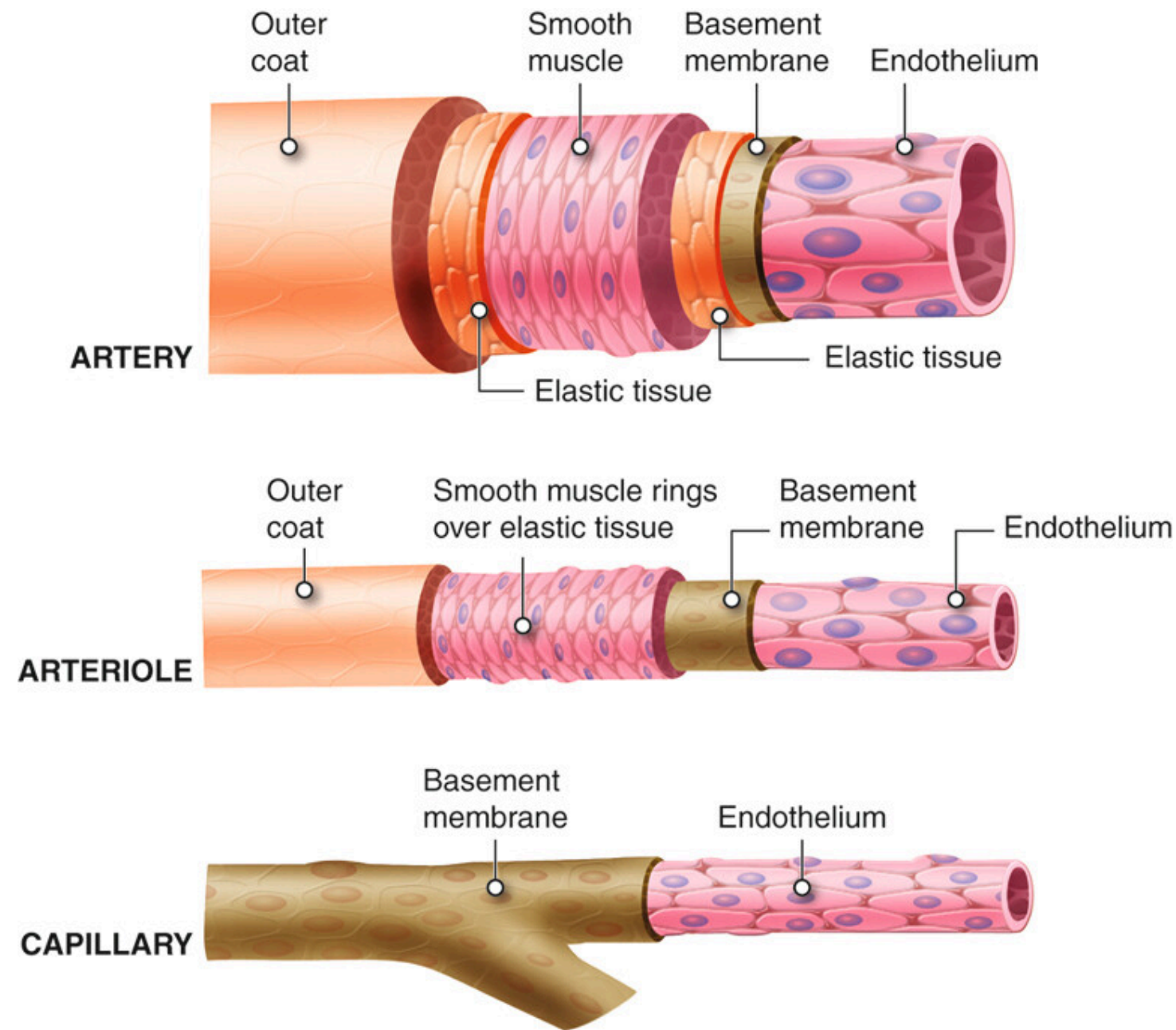
Sawdon, M. (2010).

Skeletal Muscle Circulation

- Blood flow to skeletal muscle varies enormously depending on activity:
 - Resting muscle: receives about 15–20% of cardiac output.
 - During intense exercise: may receive up to 80–90% of cardiac output.
- This circulation must balance **metabolic demand, temperature regulation, and systemic pressure maintenance.**



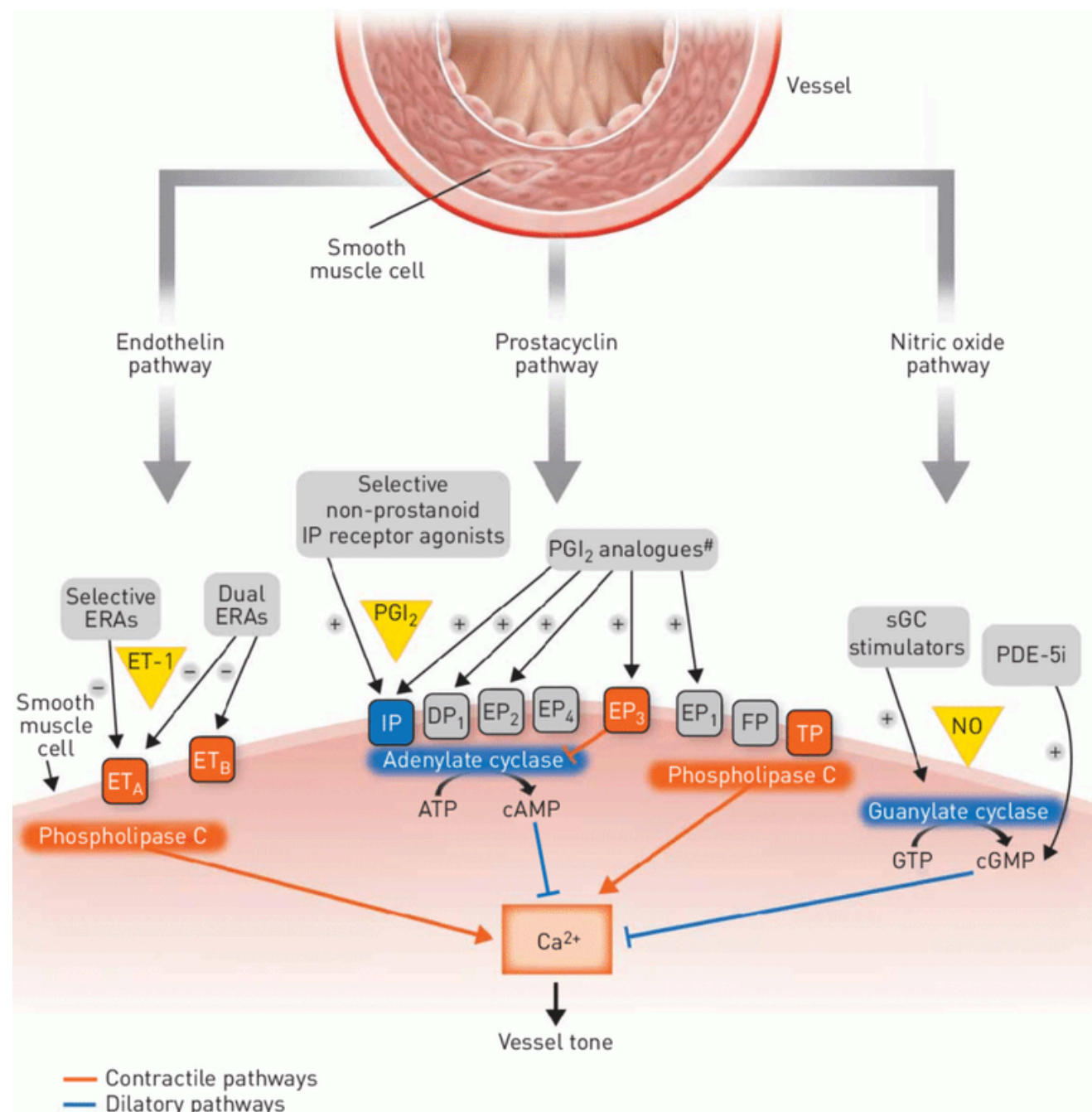
Control of Skeletal Muscle Blood Flow



1. At Rest – Neural (Extrinsic) Control Dominates

- Sympathetic vasoconstrictor fibres maintain high vascular tone.
- **Noradrenaline** acts on **α_1 -adrenergic receptors** in vascular smooth muscle → vasoconstriction.
 - This mechanism helps sustain arterial blood pressure and redistribute flow to vital organs when muscle demand is low.
- **Sympathetic cholinergic vasodilator fibres** (less prominent in humans) may participate during emotional stress (“fight-or-flight” response).

Control of Skeletal Muscle Blood Flow



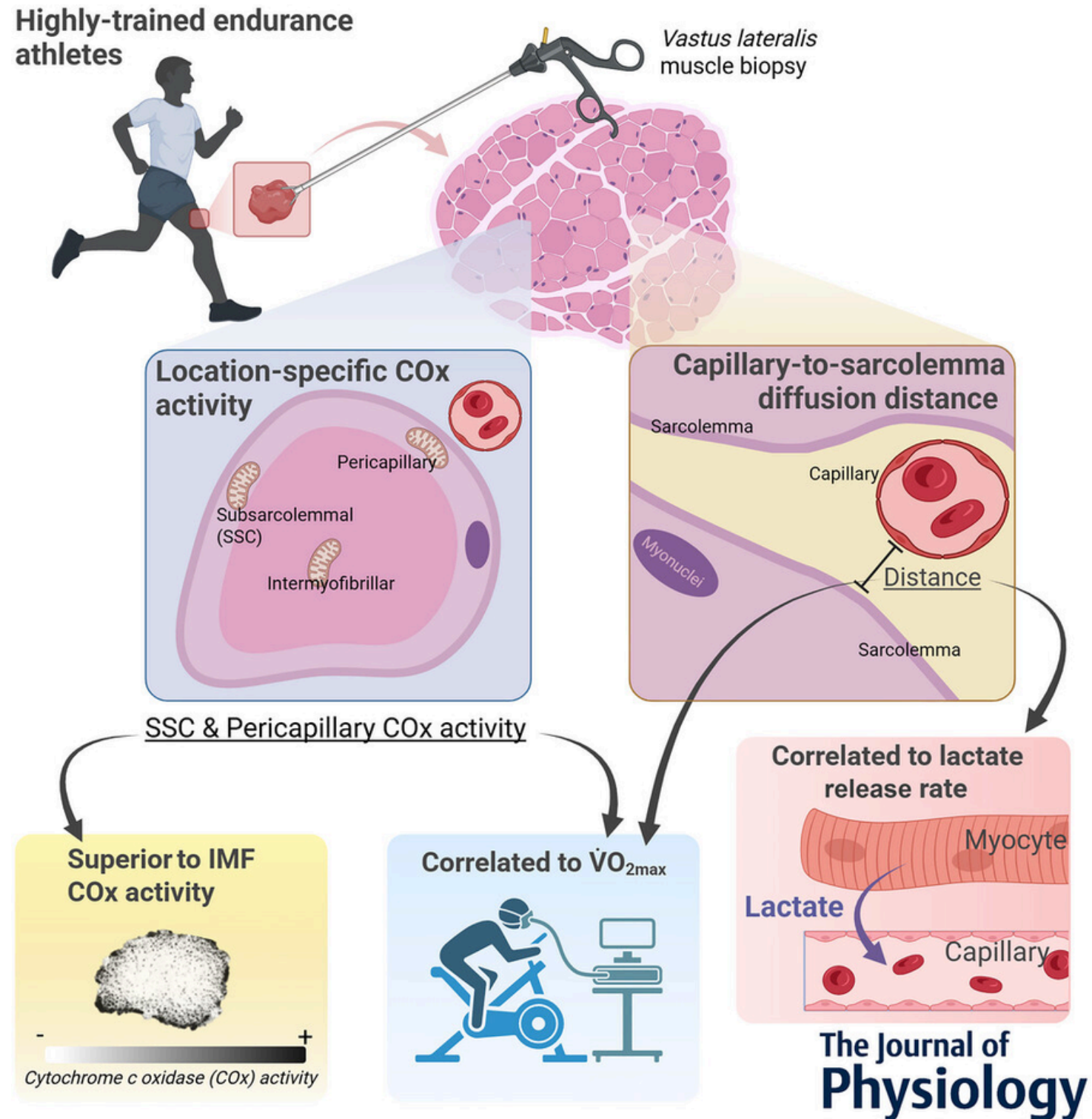
2. During Exercise – **Local (Intrinsic) Metabolic Control** Dominates

- As metabolic activity rises, vasodilator metabolites override sympathetic tone → **active hyperaemia**.
- Major local vasodilators:
 - Adenosine
 - Potassium ions (K⁺) (released during repeated depolarization)
 - Carbon dioxide (CO₂)
 - Lactic acid (H⁺)
 - Decreased oxygen tension (hypoxia)
- These act directly on arterioles or via the endothelium (releasing **NO**, **prostacyclin**) → relaxation of vascular smooth muscle.

3. Reactive Hyperaemia

- After temporary occlusion of blood flow (e.g., during muscle contraction or external compression), flow increases markedly when perfusion resumes.
- This is due to accumulation of local metabolites and oxygen debt repayment.

Special Features



Capillary recruitment:

- Resting muscle perfuses only a fraction of its capillaries.
- During exercise, capillary recruitment improves diffusion distance and oxygen exchange efficiency.

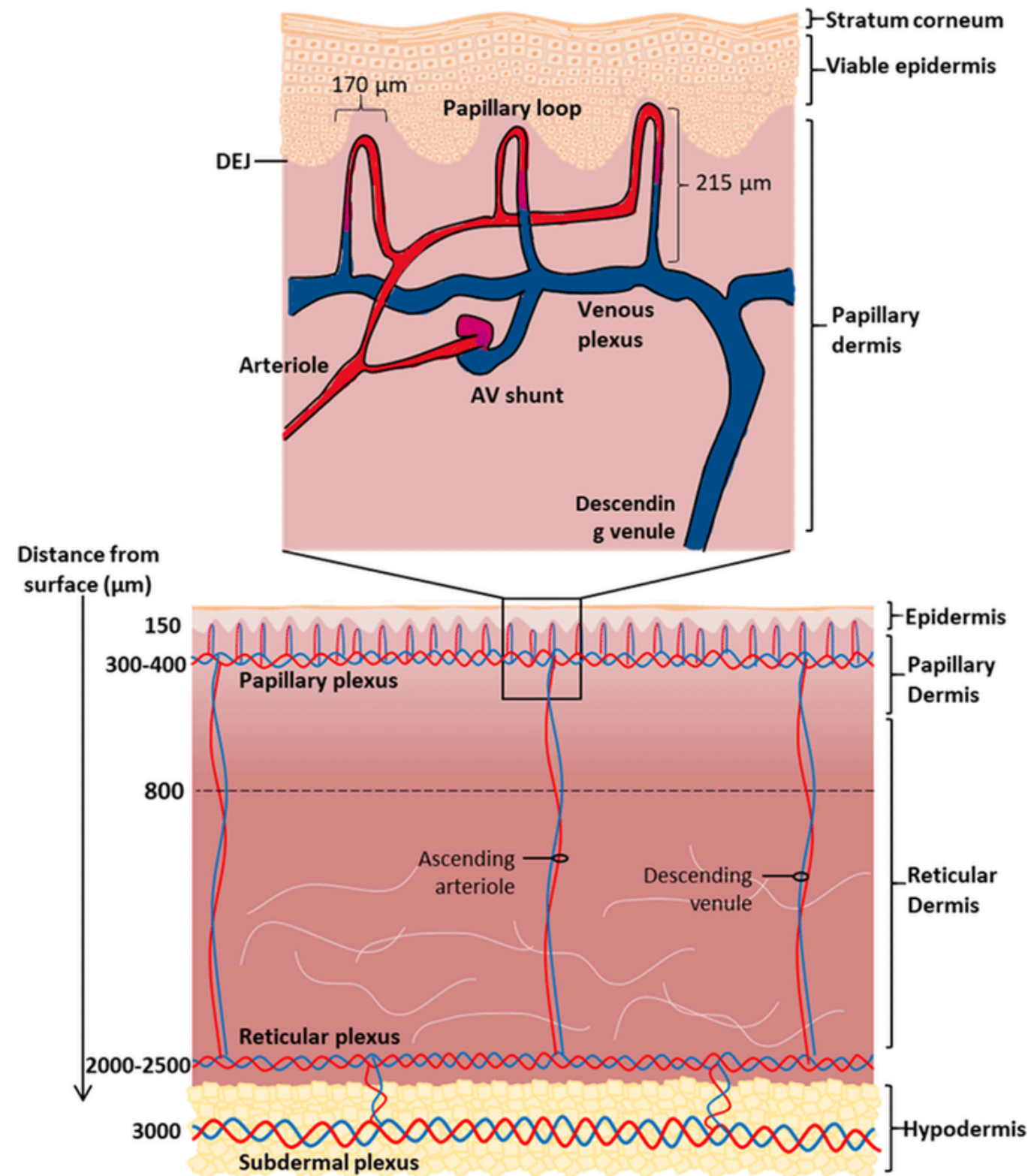
Muscle pump effect:

- Rhythmic contraction of skeletal muscle compresses veins, promoting venous return to the heart (important during exercise).

Sympathetic restraint:

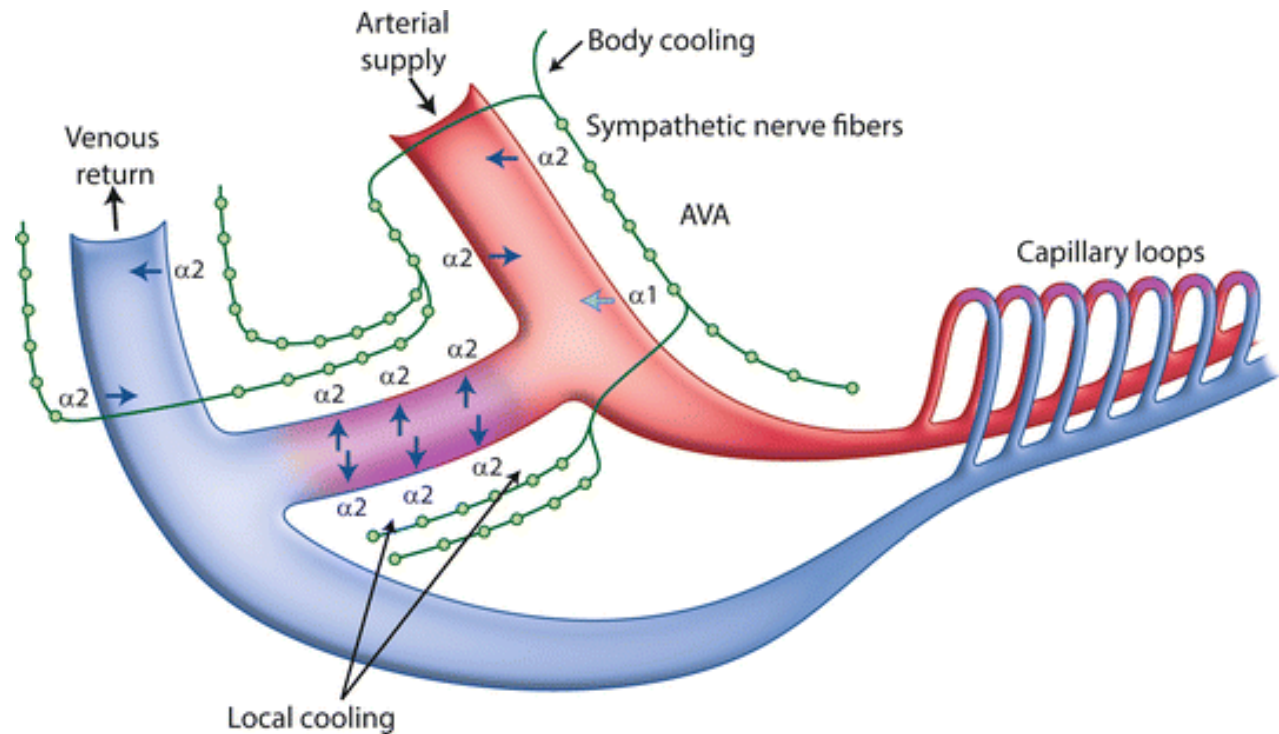
- During exercise, although sympathetic activity increases systemically, metabolic vasodilation in active muscle dominates locally (“**functional sympatholysis**”).

Cutaneous circulation



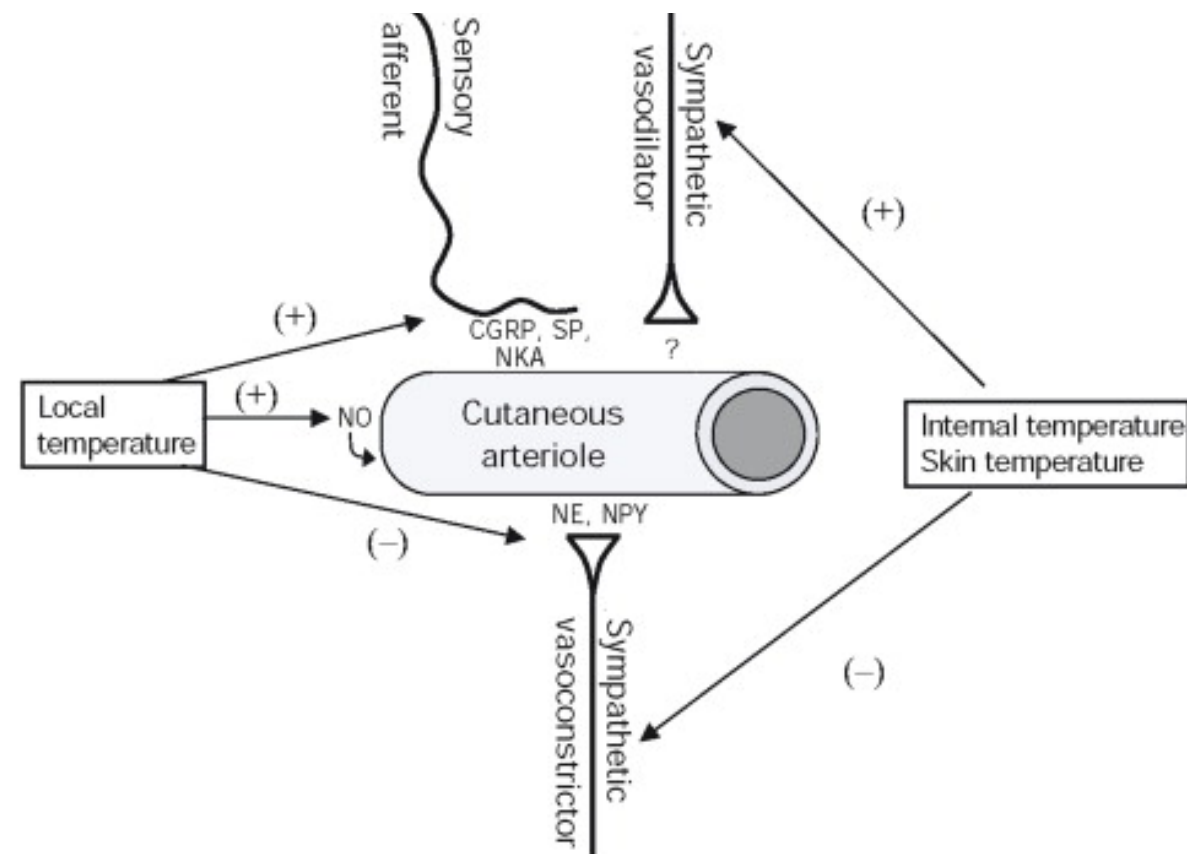
- The primary function of cutaneous circulation is **thermoregulation**, not metabolism.
- The skin receives about 5–10% of cardiac output at rest, but this can vary widely:
 - In heat: blood flow may increase up to 10-fold (vasodilation).
 - In cold: blood flow may fall to near zero (vasoconstriction).
- Metabolic demand of skin tissue is low, so blood flow changes are mainly driven by temperature control, not oxygen requirement.

Control of Cutaneous Blood Flow

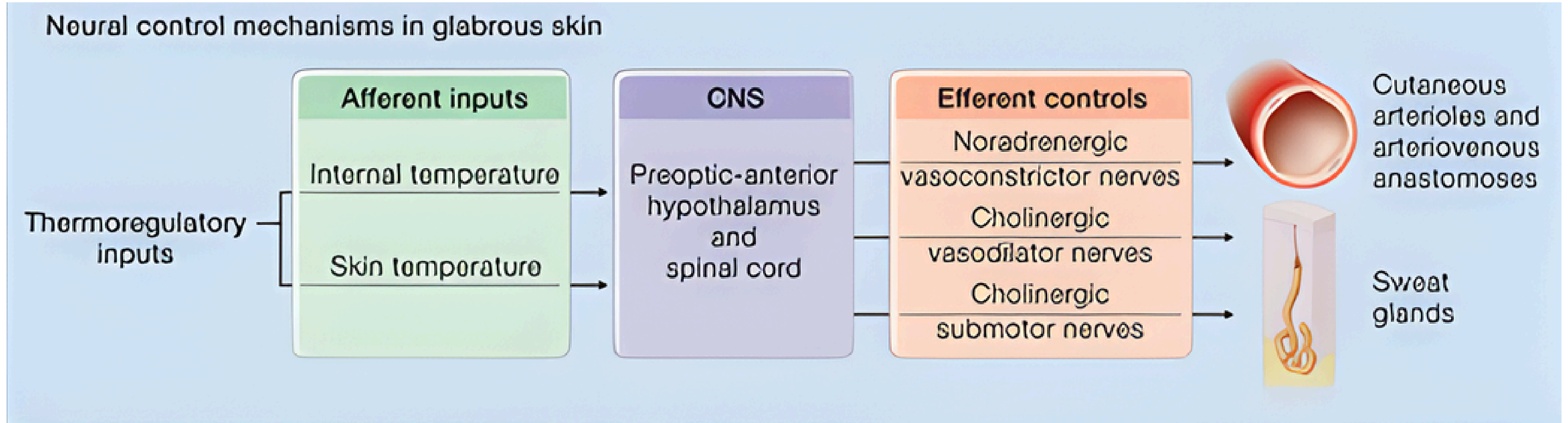


1. Neural (Extrinsic) Control – Dominant

- Sympathetic adrenergic fibres maintain tonic vasoconstrictor tone via noradrenaline (α_1 -receptors).
- Cutaneous vasodilation during heat stress is due to:
 - **Inhibition of sympathetic vasoconstrictor** activity, and
 - **Activation of sympathetic cholinergic vasodilator** fibres, which co-release acetylcholine and vasodilator substances (e.g. bradykinin) from sweat glands.



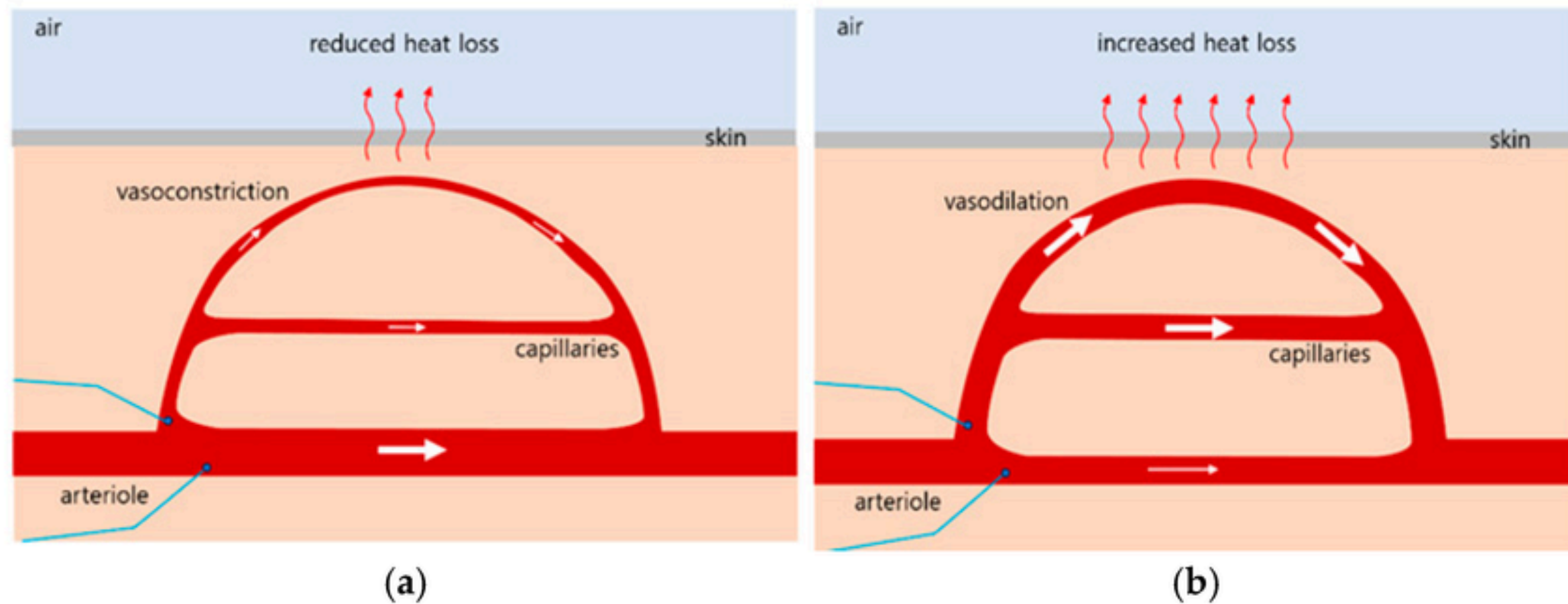
Control of Cutaneous Blood Flow



Source: Goldsmith LA, Katz SI, Gilchrist BA, Pomeroy AS, Lefkowitz DJ, Wolff K: *Rupprecht's Dermatology in General Medicine*, 8th Edition: www.accessmedicine.com

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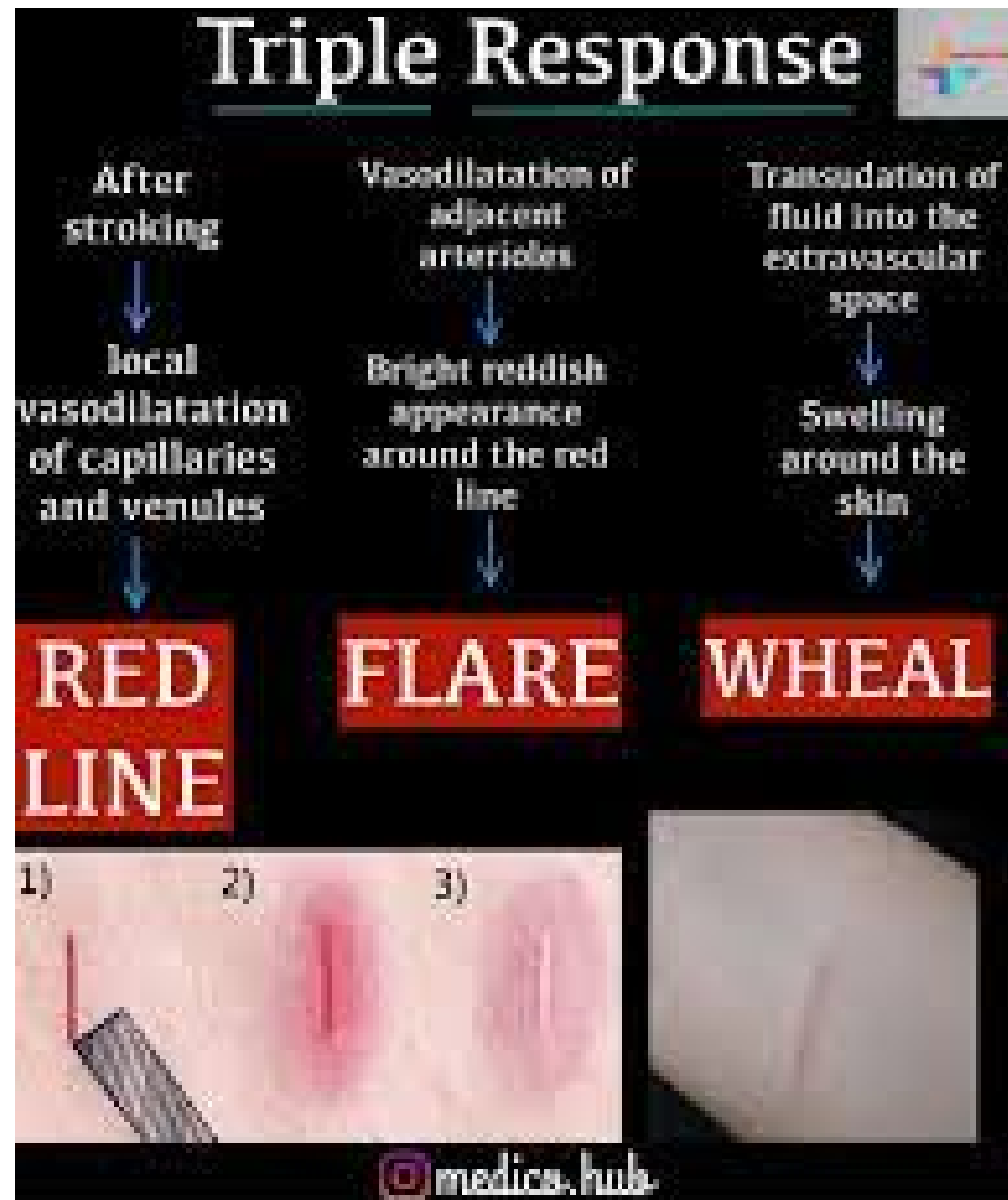
Control of Cutaneous Blood Flow



2. Local (Intrinsic) Mechanisms

- Temperature (direct effect):
 - Local warming → vasodilation of arterioles.
 - Local cooling → vasoconstriction.
- Reactive hyperaemia:
 - After brief occlusion or cold-induced vasospasm, vasodilation occurs due to accumulation of local metabolites.
- Endothelial control:
 - Nitric oxide (NO) contributes to vasodilation, particularly during heat exposure.

Control of Cutaneous Blood Flow



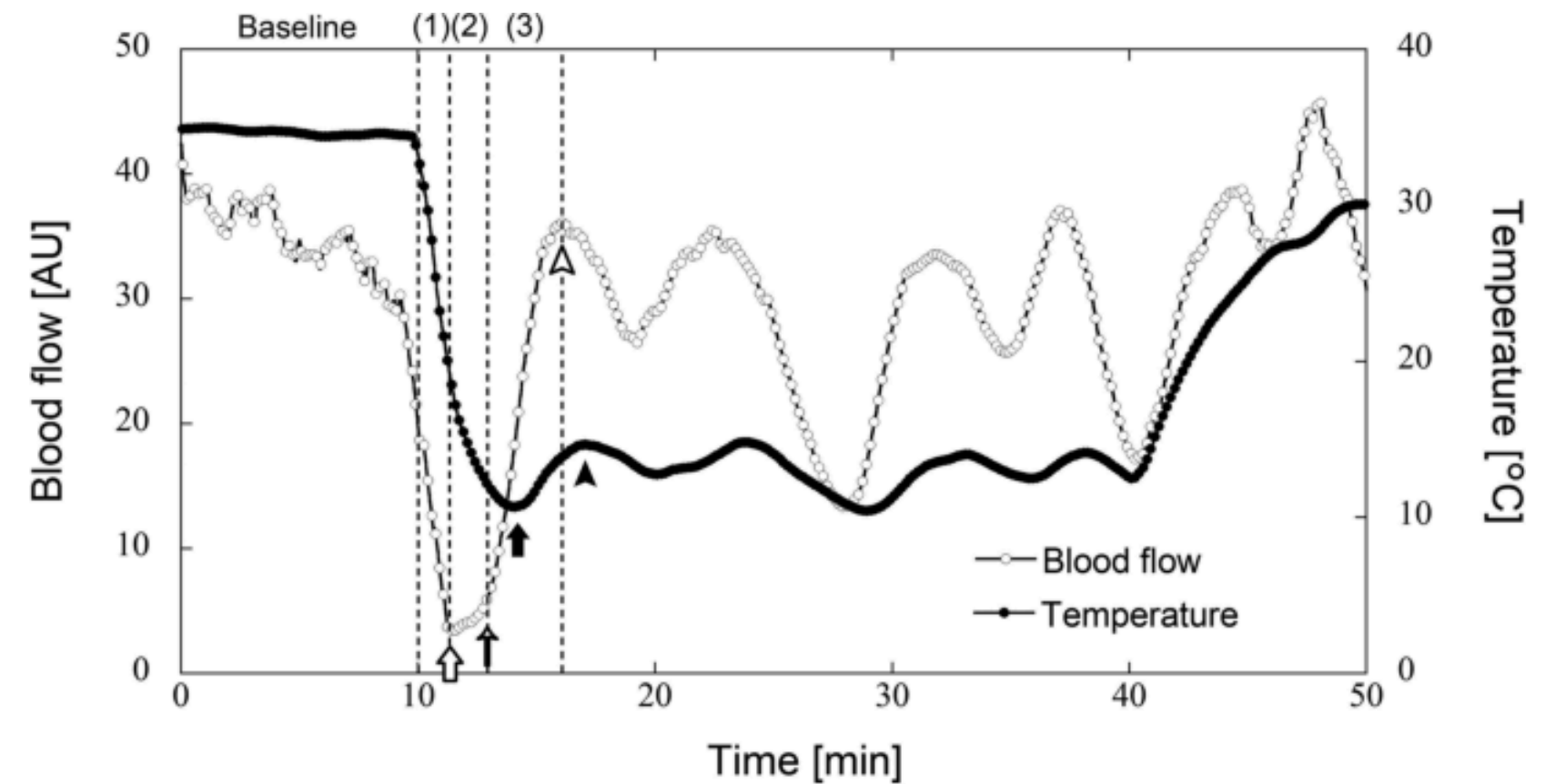
3. Special Responses

- Triple response (Lewis response):
 - Occurs after mechanical injury to skin (e.g., scratching).
 - Sequence: redness (local dilation) → flare (axon reflex dilation) → wheal (local edema).
 - Mediated by **histamine and sensory nerve reflexes**.

Control of Cutaneous Blood Flow

3. Special Responses

- Cold-induced vasodilation (“hunting reaction”):
 - Periodic vasodilation after prolonged cold exposure **to prevent tissue damage (frostbite).**
 - Common in fingers, toes, ears.



(Yasukochi et al., 2023)